

Attitude estimation of uncontrolled space objects: A Bayesian-informed swarm intelligence approach

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Abstract

The increasing congestion of Earth's orbital environment necessitates advancements in traditional Space Surveillance and Tracking (SST) methods to ensure the safe and sustainable use of space. In this context, accurately estimating the attitude of uncontrolled space objects is essential for developing effective space debris mitigation strategies and improving key predictions, such as atmospheric re-entries and collision probabilities. This study introduces the AISwarm-UKF method, a novel approach for attitude estimation of uncontrolled space objects with known geometric and optical characteristics using light curve data. The method integrates different estimation, optimisation and data analysis techniques, namely Adaptive Importance Sampling (AIS), Systematic Resampling, Particle Swarm Optimisation (PSO), Clustering and the Unscented Kalman Filter (UKF), to improve the performance of the Bayesian inference process. Applied to a realistic operational scenario, the AISwarm-UKF method demonstrates high accuracy, robustness, and computational efficiency, offering a viable solution for space situational awareness.

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1. Introduction

The unprecedented rise in satellite launches in recent years has significantly increased space debris. As orbital congestion grows, it poses serious risks to the safety of operational spacecraft and future missions, making the adoption of effective mitigation strategies essential to ensuring the long-term sustainability of the space environment.

Accurate attitude information of space objects plays a fundamental role in this effort, as it improves orbit determination accuracy, refines collision probability and atmospheric re-entry predictions, and aids in detecting anomalies in both active and inactive satellites. Moreover,

the design of some space debris mitigation missions, such as Active Debris Removal (ADR) and In-Orbit Servicing (IOS), relies on precise attitude knowledge to ensure operational effectiveness and minimise the risk of generating additional debris.

The attitude of a space object can be estimated by analysing the sunlight reflected from its surfaces and captured by ground-based optical sensors. The temporal variation of these photometric intensity measurements forms a light curve, and the process of deriving an object's unknown parameters from it is traditionally referred to as the light curve inversion problem.

1.1. State of the art

Originally, light curve inversion methods were developed to estimate the shape and rotational state of asteroids.

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The application of these techniques to artificial objects in Earth orbit was first investigated by researchers at the Air Force Maui Optical Station (AMOS) in the early years of the 21st century (Hall et al., 2006; Hall et al., 2007).

Hall et al. (2007) were the first to describe the challenges of applying light curve inversion to artificial objects. The primary difficulties arise from the large number of parameters needed to characterise the complex, highly angular shapes of satellites, whose surfaces may consist of diverse materials that reflect sunlight both diffusely and specularly. Additionally, the highly non-linear evolution of attitude over an observation period further complicates efficient computational determination.

To address these challenges, Hall et al. (2007) proposed decoupling the simultaneous attitude and shape estimation problem for space objects into attitude-independent shape analysis and shape-independent attitude analysis, a strategy that has since been widely adopted. In that study, several shape-independent attitude estimation methods were introduced, including the use of variations in the apparent rotation rate (as observed from the sensor's perspective, that is, measured directly from the light curve) to determine the sidereal spin rate and spin axis orientation of stably rotating objects.

The attitude-independent shape estimation problem has been extensively studied since then, with the determination of the Extended Gaussian Image (EGI) and the subsequent solution to the Minkowski problem emerging as one of the most promising approaches, as demonstrated in the work of Fan and Frueh (2020). However, the dependence on prior attitude knowledge for shape estimation limits its operational applicability. Consequently, this research focuses on the attitude estimation of uncontrolled space objects. The following literature review is centred on this topic, although some of the methods discussed also attempt to estimate the object's shape and/or optical properties simultaneously.

The use of light curves for sequential attitude estimation was first explored by Wetterer and Jah (2009). In their work, the attitude of a simulated rocket body was estimated using an Unscented Kalman Filter (UKF) and assuming precise knowledge of its geometric and surface optical properties. The study evaluated the convergence of the UKF by varying initial attitude estimates within a 3σ range of the true values. The UKF successfully recovered the correct attitude parameters across these variations. However, it was also observed that when initial attitude offsets were excessively large, the UKF struggled to converge to the correct solution. This difficulty arises from the symmetry of the object, which can produce nearly identical light curves for different rotation axis orientations.

Linares et al. (2010) extended the application of the UKF by proposing a method for the simultaneous estimation of a space object's rotational and translational states, as well as its shape. This approach employed a Multiple-Model Adaptive Estimation (MMAE) technique, which utilises a bank of UKFs, each associated with a different

candidate shape model. The overall state estimate was computed as a weighted sum of the estimates from each individual filter. Furthermore, Rush et al. (2020) introduced two distinct UKF formulations: one for scenarios in which the object rotates about a fixed axis, and another for tumbling rotational states. Their results demonstrated the challenges inherent in the light curve inversion problem, particularly the existence of multiple attitude states capable of producing equivalent light curves.

Bayesian sequential estimation methods have also been explored for attitude estimation from light curves. The most extensively studied approach is the particle filter, an approximate Bayesian estimator that uses a weighted set of samples to represent probability densities. The particle filter offers significant advantages over the UKF, particularly in its ability to handle non-Gaussian and multimodal probability density functions. These complexities often arise from the highly non-linear measurement model, which must account for specular reflection, and the potential existence of multiple equivalent solutions due to the intrinsic coupling between geometric, optical and attitude parameters.

Holzinger et al. (2012); Holzinger et al. (2014) were the first to apply a particle filter to the light curve inversion problem. They employed a Sample Importance Resampling (SIR) particle filter to estimate the attitude, even in the presence of shape model uncertainty, and to track actively manoeuvring satellites. Linares et al. (2014) also explored the use of a SIR particle filter for the estimation of orientation and angular rate of uncontrolled space objects, assuming a known shape model and using Generalised Rodrigues Parameters (GRPs) for local error representation. In addition, Bernard (2021) compared the performance of the UKF and the SIR particle filter, concluding that the latter provided the most robust results.

Variations of the SIR particle filter have also been investigated, either independently or in combination with other estimation techniques. Coder et al. (2015); Coder et al. (2018); Coder et al. (2018) employed the Rao-Blackwellised Particle Filter (RBPF) to maintain computational efficiency when estimating a large number of states. The RBPF uses a particle-based representation for the non-linear states, with each particle associated with a multivariate Gaussian distribution for the remaining linear states, which are estimated using a Kalman filter. Du et al. (2018) used an Unscented Particle Filter (UPF) to estimate the attitude of geostationary satellites, integrating the UKF to refine particle selection by redesigning the importance density function. The RMS-UKF was also employed to partially correct the prediction covariance matrix, addressing the limitations of the UKF and mitigating particle degeneracy. Yun and Zanetti (2020) proposed a particle filter augmented with an expectation–maximisation clustering algorithm to approximate the state distribution using Gaussian mixture models. This approach enabled the simultaneous estimation of position, velocity, attitude, angular rates, and surface parameters of a space object in

near-geostationary orbit. More recently, Wang et al. (2023) combined two particle filters to estimate the attitude of geostationary objects. In this method, the first particle filter identifies particles near the true state, thereby allowing the second particle filter to start with a better initial distribution of particles.

Alternative methods to the particle filter have also been explored. For example, Linares and Crassidis (2018) investigated an Adaptive Hamiltonian Markov Chain Monte Carlo (AHMC) approach for the simultaneous estimation of an object's shape, attitude, and surface parameters. This method generates samples from the posterior probability density function using relatively simple proposal and rejection rules. Cabrera et al. (2023) proposed the Adaptive Gaussian Mixtures Unscented Kalman Filter (AGMUKF), in which the state probability density function is represented as a Gaussian mixture, with the UKF used to propagate and update each Gaussian kernel. In a more recent study, Hara et al. (2024) combined the UKF with a non-parametric observation model constructed using Gaussian Process Regression (GPR), a machine learning technique. This hybrid approach improves the robustness of attitude estimation, especially in the presence of unknown surface parameters, and has been applied to space objects in geosynchronous orbits.

Finally, an alternative approach to addressing the light curve inversion problem involves the use of optimisation algorithms. Clark et al. (2020) assessed the performance of five different methods, including Particle Swarm Optimisation (PSO), for estimating the shape, attitude, and optical properties of space objects. PSO is also employed in Clark et al. (2022) to evaluate two different observation models. Burton et al. (2024) applied a combination of two PSO executions to estimate arbitrary torque-free attitude motion, assuming known shape and reflective properties of the object. In this approach, the first PSO identifies a set of attitudes capable of reproducing the measurement at the initial time, while the second PSO refines the search over the space of initial attitudes and angular velocities to determine the initial state that best reproduces the entire light curve.

1.2. Research objectives

Before describing the methodology used in this work to address the light curve inversion problem for attitude estimation, it is first necessary to explain the research objectives and justify the selected approach based on insights from the literature review presented above.

The primary objective of this work is to estimate the attitude of uncontrolled space objects—those lacking active control or stabilisation mechanisms—with known shape and surface optical properties. This is achieved by using photometric measurements from one or more light curves obtained by ground-based optical sensors. The

methodology aims to address the most challenging scenario, in which photometric measurements are available from only a single sensor.

The space object's attitude is estimated at a specific moment in time, assuming a worst-case scenario in which no prior knowledge of its attitude is available. This objective is analogous to that of Initial Orbit Determination (IOD) (Escobal, 1965), where an unknown orbit is estimated using sensor measurements (e.g. range and range rate) at a given time, typically that of the most recent observation.

Therefore, the proposed attitude estimation method can be applied to operational use cases such as monitoring the attitude of known satellites to detect anomalies—e.g. those caused by non-catastrophic collisions with other space objects—or planning Close Proximity Operations (CPOs) for space debris mitigation, including Active Debris Removal (ADR) and In-Orbit Servicing (IOS) missions.

An important conclusion drawn from the literature review presented in Section 1.1 is that the light curve inversion problem is significantly affected by measurement ambiguities, that is, the existence of multiple valid solutions that produce the same photometric signature. These ambiguities can arise for two main reasons:

- The symmetry in the space object's geometry, combined with nearly identical optical properties of its symmetric surfaces. This type of ambiguity cannot be resolved when the object's properties are perfectly symmetric, so a robust methodology must identify and accurately estimate all valid solutions. A notable example of space debris likely to exhibit this type of ambiguity is rocket bodies.
- The inherent coupling between attitude, object geometry, and surface optical properties, meaning that different combinations of these parameters can produce equivalent effects on the resultant light curve. This type of ambiguity is difficult to resolve when only photometric measurements from a single sensor are available. As with the previous case, a robust methodology must identify all valid solutions. The possibility of resolving this ambiguity is demonstrated in Section 3 using stereoscopic measurements, i.e. photometric measurements taken by sensors at different locations within approximately the same observation interval.

In addition to addressing the challenge of measurement ambiguity, the proposed methodology is designed based on insights from various estimation techniques reviewed in Section 1.1, particularly the UKF and the particle filter in their different variants. Consequently, the light curve inversion method for attitude estimation introduced in this work has been developed according to the following key principles:

1. Exploration of the complete solution space

As outlined earlier, no a priori knowledge of the space object's attitude is assumed, requiring consideration of the complete solution space. Moreover, the methodology must be robust enough to identify all the modes of the posterior probability density function, that is, all valid solutions that may arise due to measurement ambiguity.

Therefore, methods that rely on an initial guess, such as the UKF, are not well suited for this application. This limitation is clearly illustrated in Fig. 1, which presents a statistical analysis of the UKF's performance in attitude estimation based on a dataset of 1000 synthetic light curves. A detailed discussion is provided in Section 2.7.

2. Batch processing of photometric measurements

Particle filtering is another well-established estimation technique in the literature. Unlike the UKF, the particle filter does not require an accurate initial estimate, as it can be initialised with a uniform distribution of particles. However, despite its effectiveness for real-time operations, particle filtering is a sequential estimation method. This makes it less suitable for the current application, where photometric measurements are not continuously available and instead depend on the space object's visibility from the sensor.

In general, each visibility interval is relatively short compared to the object's orbital period, with near real-time (albeit offline) processing of the complete light curve taking place once the visibility interval has concluded. Therefore, the proposed method adopts a batch-processing strategy for the photometric measurements, enabling full exploitation of the available light curve data. This approach not only reduces the risk of convergence to locally optimal solutions but also improves computational efficiency by facilitating parallelisation of the calculations.

3. Efficient convergence to accurate solutions

Previous studies (Wang et al., 2023; Burton et al., 2024) have proposed a two-step estimation process, where the first step identifies potential candidate solutions and the second step refines them to determine the correct solution. This strategy can be computationally expensive, as it either requires processing a large number of measurements in the first step to minimise the presence of local optima or, if fewer measurements are used in the first step, results in an excessive number of candidate solutions needing further analysis in the second step.

To maximise computational efficiency, the proposed attitude estimation method simultaneously explores the solution space and refines the estimation of promising local optima. The approach is iterative, with early iterations focusing on exploring high-probability density regions, while later iterations prioritise convergence and the refinement of the most viable candidate solutions.

2. Methods

This section describes the proposed light curve inversion method for attitude estimation, named AISwarm-UKF, which results from the coordinated application of different estimation, optimisation, and data analysis techniques. The method is designed with operational feasibility in mind, ensuring both accuracy and robustness while maintaining computational efficiency.

To achieve these objectives, the AISwarm-UKF method integrates Adaptive Importance Sampling (AIS), Systematic Resampling, Particle Swarm Optimisation (PSO), Clustering, and the Unscented Kalman Filter (UKF). The underlying principles of each method are

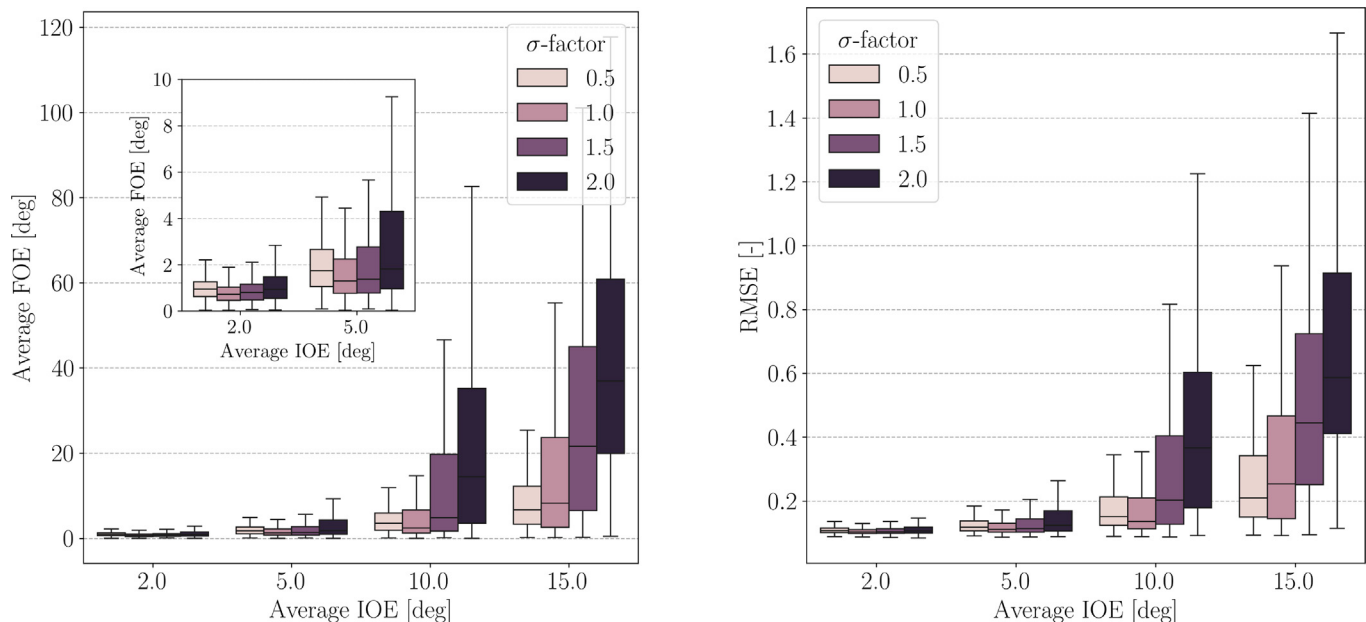


Fig. 1. Statistical analysis of the UKF's performance in attitude estimation.

summarised below, followed by a comprehensive explanation of their integration for effective attitude estimation.

Before discussing the details of the AISwarm-UKF method, the attitude and observation models required to simulate the light curves produced by different attitude states are presented in Sections 2.1 and 2.2, respectively.

2.1. Attitude model

In general, the attitude motion of an uncontrolled space object exhibits tumbling, a complex rotational state in which the axis of rotation changes over time. This motion can be described as a combination of spinning, precession and nutation. Spinning refers to rotation about an instantaneous axis and remains stable in free space when the spin axis is aligned with the object's principal axis of maximum inertia (Schaub and Junkins, 2009). Precession and nutation arise when the object does not rotate about a principal axis of inertia. Precession is characterised by the spin axis describing a conical trajectory around a fixed reference axis. Nutation is a periodic oscillation of the spin axis superimposed on precession, usually resulting from uneven interactions between external torques and objects with asymmetric shapes or non-uniform mass distributions.

In this study, the evolution of the attitude state during the period in which the space object is visible from the ground station is modelled purely kinematically, neglecting the effect of external torques. This simplification is justified, as perturbing torques arising from sources such as the non-uniform gravitational field, eddy currents, solar radiation pressure and atmospheric drag, are negligible over the duration of the visibility interval. In fact, even if the external torques were accounted for, the accuracy of the estimation would see little improvement in practical scenarios, since key parameters influencing the dynamics, including the eddy currents tensor and mass distribution, are generally unknown.

Considering a body-fixed reference frame B , attached to the space object, and an inertial reference frame I , such as the Geocentric Celestial Reference Frame (GCRF) (Kaplan, 2005), the kinematic equation governing the rotational motion of a rigid body is given by Wertz (1978):

$$\dot{\mathbf{q}}_B^I = \frac{1}{2} \mathbf{q}_B^I \otimes \boldsymbol{\omega}_{q,B/I} = \frac{1}{2} \boldsymbol{\Omega} \mathbf{q}_B^I \quad (1)$$

$$\boldsymbol{\Omega} = \boldsymbol{\Omega}(\boldsymbol{\omega}_{q,B/I}) = \begin{bmatrix} 0 & -\omega_x & -\omega_y & -\omega_z \\ \omega_x & 0 & \omega_z & -\omega_y \\ \omega_y & -\omega_z & 0 & \omega_x \\ \omega_z & \omega_y & -\omega_x & 0 \end{bmatrix} \quad (2)$$

where $\mathbf{q}_B^I$ is the quaternion representing the rotation of frame B relative to frame I at some time t , and $\boldsymbol{\omega}_{q,B/I} = [0, \boldsymbol{\omega}_{B/I}]^T = [0, \omega_x, \omega_y, \omega_z]^T$ is the quaternion repre-

sensation of the instantaneous angular velocity of frame B relative to frame I , expressed in the rotating frame B .

However, a complete characterisation of a tumbling attitude state—which would require accounting for a time-varying angular velocity $\boldsymbol{\omega}_{B/I}$ —is generally unnecessary in operational contexts. This is because most space objects that could be targeted for attitude estimation typically satisfy one or both of the following conditions:

- Over extended periods, energy dissipation causes space objects that have remained free from collisions or explosions to settle into a flat spin, rotating solely around their principal axis of maximum inertia. This phenomenon is caused by dissipative torques like those generated by eddy currents and atmospheric drag. Indeed, even in the absence of external torques, objects gradually lose energy through internal damping because they are not perfectly rigid (Schaub and Junkins, 2009).
- Although some space objects may undergo precession, the precession period is generally much longer than the spin period—typically on the order of tens of seconds for spinning, compared to hundreds or even thousands of seconds for precession (Fu et al., 2010). Considering that typical visibility intervals for Low Earth Orbit (LEO) objects are around five minutes, the precession period usually extends beyond the observation window. Consequently, sensors primarily capture the photometric signature induced by the spinning motion.

As a result, the kinematic attitude model used to compute the orientation of the space object at the times of the actual photometric measurements assumes a purely spinning motion about a fixed axis. Consequently, the components of the angular velocity $\boldsymbol{\omega}_{B/I}$ remain constant, the matrix $\boldsymbol{\Omega}$ is also constant, and Eq. (1) can be readily integrated to yield the following closed-form solution for the kinematic equation (Wertz, 1978):

$$\begin{aligned} \mathbf{q}_B^I(t) &= \exp\left(\frac{1}{2} \boldsymbol{\Omega} t\right) \mathbf{q}_{B,0}^I \\ &= \left[\cos\left(\frac{\omega_{B/I}}{2} t\right) \mathbf{I} + \frac{1}{\omega_{B/I}} \sin\left(\frac{\omega_{B/I}}{2} t\right) \boldsymbol{\Omega} \right] \mathbf{q}_{B,0}^I \end{aligned} \quad (3)$$

Therefore, the kinematic attitude model is defined by the orientation of the space object at a reference time $\mathbf{q}_{B,0}^I$ and its constant spin angular velocity $\boldsymbol{\omega}_{B/I}$, resulting in six unknown parameters to be estimated. The attitude parameters selected to address the light curve inversion problem in this work are:

- The orientation of the body-fixed frame B with respect to the inertial frame I , represented by a quaternion or Euler angles (yaw, pitch and roll) following the ZYX convention.
- The orientation of the spin axis relative to frame B , expressed in terms of azimuth and elevation angles.
- The spin period.

2.2. Observation model

Solving the light curve inversion problem for attitude estimation requires a measurement model to simulate photometric observations and evaluate different attitude states. In this work, the **Grial** tool from GMV is used. **Grial** is a high-fidelity light curve simulator that accurately computes the irradiance reaching the optical sensor after reflection from each illuminated and visible surface of a three-dimensional space object. The optical properties of each surface can be individually specified, and self-shadowing interactions between different parts of the object are accounted for, thereby enabling realistic simulations of photometric measurements.

Grial uses OpenGL to graphically compute the contribution of each illuminated and visible pixel on the surface of the space object to the reflected light. OpenGL allows these calculations to be performed on a computer's Graphics Processing Unit (GPU), achieving significantly faster processing speeds compared to the Central Processing Unit (CPU). On average, **Grial** simulates an accurate photometric measurement in just 1.4 ms. This high-performance capability is crucial for ensuring the operational viability of the proposed AISwarm-UKF attitude estimation method.

The main characteristics of the observation model implemented in **Grial** are described below, following the sequential process: first, the incidence of sunlight on the space object; then, the reflection of light from its surfaces; and finally, the reception of the reflected light by a ground-based optical sensor.

2.2.1. Incident irradiance

The irradiance or radiant flux (power per unit area) received from the Sun by the space object can be computed as the ratio of total radiated power by the Sun and the area over which the radiated power is distributed, which yields the well-known inverse square law for light (Carroll and Ostlie, 2017):

$$I_i = \frac{L_\odot}{4\pi r_\odot^2} \beta \quad (4)$$

where the radiant power emitted by the Sun, also called solar luminosity, is $L_\odot = 3.839 \times 10^{26}$ W. The area of a sphere surrounding the Sun on which surface is the space object is $A = 4\pi r_\odot^2$, where $r_\odot \approx 1$ AU is the distance between the Sun and the space object.

A factor β has been introduced in Eq. (4) to consider the partial occlusion of sunlight by the Earth. The fraction of sunlight blocked by the Earth is computed considering the Sun-Earth-object relative geometry, modelling the Sun as a sphere of radius $R_\odot$ and the Earth as a one-axis ellipsoid (Parraud and Maisonobe, 2025):

$$\beta = \min \left\{ 0, \max \left\{ \frac{\alpha + \alpha_\odot - \alpha_\oplus}{2\alpha_\odot}, 1 \right\} \right\} \quad (5)$$

where $\alpha_\odot$ and $\alpha_\oplus$ are the apparent radius of the Sun and the Earth, respectively, as seen from the space object, and α is the angle between the Earth's centre and the Sun's centre as seen from the space object, which can be computed as follows:

$$\alpha_\odot = \arcsin \left(\frac{R_\odot}{r_\odot} \right) \quad (6)$$

$$\alpha_\oplus = \arccos \left(\frac{\mathbf{r}_\oplus \cdot \mathbf{r}_{\text{limb}}}{r_\oplus r_{\text{limb}}} \right) \quad (7)$$

$$\alpha = \arccos \left(\frac{\mathbf{r}_\oplus \cdot \mathbf{r}_\odot}{r_\oplus r_\odot} \right) \quad (8)$$

where $\mathbf{r}_\oplus$ is the position vector from the space object to Earth's centre and $\mathbf{r}_{\text{limb}}$ is the vector from the space object to Earth's limb point (tangent point of Earth's ellipsoid).

2.2.2. Reflected irradiance

A reflectance model is employed to predict the directional distribution of sunlight reflected by the space object, which is essential for calculating the intensity of light reaching the ground-based sensor.

The space object is modelled as a collection of facets, each defined by its surface area, normal vector, and scattering properties. The light reflected by each facet depends on the orientation of the facet relative to the Sun and the sensor. The total light reflected by the space object is obtained by aggregating the contributions from all the facets that are both illuminated by the Sun and visible from the sensor.

The light reflection geometry is defined by the unit normal to the facet $\mathbf{n}$, the unit vectors in the direction of the light source $\mathbf{l}$ and the sensor $\mathbf{v}$, and the unit normal to a hypothetical surface that would reflect light specularly from the light source to the viewer $\mathbf{h}$ (in the direction of the angular bisector of $\mathbf{l}$ and $\mathbf{v}$).

The Bidirectional Reflectance Distribution Function (BRDF), f_r , characterises the reflection of light from an opaque surface. It is formally defined as the ratio of the differential reflected radiance in the outgoing direction $\mathbf{v}$ to the differential irradiance incident from the incoming direction $\mathbf{l}$ (Nicodemus, 1965). Given the large distance between the Sun and the space object, the incident solar radiation can be approximated as collimated (i.e. consisting of parallel rays) upon reaching the object's surface. Consequently, the incoming irradiance is concentrated in a single direction, and the reflected radiance results solely from that irradiance. Under this assumption, the BRDF can be expressed as:

$$f_r = \frac{L_r}{I_i} \quad (9)$$

where I_i is the irradiance incident on an object's surface, as given by Eq. (4), and L_r is the radiance reflected in the outgoing direction.

In addition, the differential reflected irradiance dI_r received by a differential solid angle $d\Omega_o$ around the outgoing direction $\mathbf{v}$ can be computed from the reflected radiance

in that direction L_r as (Ramamoorthi and Hanrahan, 2001):

$$dI_r = L_r (\mathbf{v} \cdot \mathbf{n}) d\Omega_o \quad (10)$$

Therefore, the reflected irradiance can be expressed in terms of the incident irradiance and the BRDF as follows:

$$dI_r = f_r I_i (\mathbf{v} \cdot \mathbf{n}) d\Omega_o = (f_{r,s} + f_{r,d}) I_i (\mathbf{v} \cdot \mathbf{n}) d\Omega_o \quad (11)$$

where it has already been made explicit that the bidirectional reflectance can be decomposed into two distinct components: diffuse and specular (the background or ambient illumination can be considered negligible in outer space).

The diffuse component $f_{r,d}$ represents light that is scattered approximately equally in all directions. This scattering can occur either due to internal scattering, where incident light penetrates below the material's surface, or from multiple reflections when the surface is sufficiently rough. The specular component $f_{r,s}$, on the other hand, represents light that is reflected directly from the surface of the material. This component corresponds to highlights, that is, light concentrated around the mirror reflection direction.

The selected BRDF for the specular term is that proposed by Cook and Torrance (1982), with modifications made to improve the accuracy of the simulations. Cook-Torrance's BRDF is a physically realistic model for specular reflection that adheres to energy conservation principles (Wetterer et al., 2014) and incorporates the physical characteristics of the materials on the object's surface. This model treats each surface element as a distribution of smooth microfacets. The mathematical expression for this BRDF is:

$$f_{r,s} = \frac{FDG}{4(\mathbf{v} \cdot \mathbf{n})} \quad (12)$$

where D is the facet slope distribution function, F is the Fresnel factor, and G is the geometrical attenuation factor.

The Fresnel factor provides the fraction of incident light that is reflected by microfacets whose normal vector aligns with the halfway vector direction $\mathbf{h}$, and it varies depending on the angle of incidence of the incident light and the material's properties. It can also be interpreted as the specular reflection coefficient of a material at a given halfway vector direction with respect to the specular reflection coefficient at normal incidence. The Fresnel factor is computed with Schlick's model (Schlick, 1994), which applies a rational approximation to the Cook-Torrance model to obtain a simpler and computationally efficient formula:

$$F = k_s + (1 - k_s)(1 - \mathbf{h} \cdot \mathbf{n})^5 \quad (13)$$

where k_s is the reflection coefficient for light incoming parallel to the normal, i.e. the value of the Fresnel factor that corresponds to the minimal reflection.

The facet slope distribution function D characterises the statistical distribution of microscopic facet slopes on a surface, which in turn determines the interaction of light with

the surface at a microscopic level. Specifically, it represents the statistical density of microfacets oriented in a given direction. In this case, the halfway vector is considered, as it corresponds to the orientation a microfacet normal $\mathbf{n}$ must have to reflect light specularly from the incident direction $\mathbf{l}$ towards the sensor direction $\mathbf{v}$.

The Beckmann distribution proposed by Cook and Torrance is replaced by the GGX model proposed by Walter et al. (2007), as it is more accurate for rough materials. The GGX distribution has a broader, slower-decaying tail, allowing more microfacets to contribute to reflections at grazing angles, making it more suitable for modelling surfaces with broad highlights and glancing reflections, particularly for materials like metals:

$$D = \frac{n_s^2}{\pi((\mathbf{h} \cdot \mathbf{n})^2(n_s^2 - 1) + 1)^2} \quad (14)$$

where n_s is the roughness coefficient.

The geometrical attenuation factor G determines how much light reaches or escapes from a surface based on the microfacet self-shadowing (a microfacet blocking incoming light from reaching other parts of the surface) and self-masking (a microfacet blocking outgoing light from reaching the viewer) effects:

$$G = \min \left\{ 1, \frac{2(\mathbf{h} \cdot \mathbf{n})(\mathbf{v} \cdot \mathbf{n})}{\mathbf{v} \cdot \mathbf{h}}, \frac{2(\mathbf{h} \cdot \mathbf{n})(\mathbf{l} \cdot \mathbf{n})}{\mathbf{v} \cdot \mathbf{h}} \right\} \quad (15)$$

For the diffuse component of the BRDF, the Lommel-Seeliger model (Fairbairn, 2005) has been selected over the Lambertian model proposed by Cook and Torrance. While the latter assumes uniform diffuse reflection, the Lommel-Seeliger model accounts for the angles of incidence and emission, ensuring better alignment with the physical behaviour of surfaces exposed to sunlight in space. Moreover, it assumes that incident light penetrates the surface, is attenuated by scattering and absorption, and is then scattered isotropically. The Lommel-Seeliger BRDF is given by:

$$f_{r,d} = \frac{\min\{k_d, 1 - F\}(\mathbf{l} \cdot \mathbf{n})}{2\pi(\mathbf{l} \cdot \mathbf{n} + \mathbf{v} \cdot \mathbf{n})} \quad (16)$$

where $\min\{k_d, 1 - F\}$ defines the diffuse reflection coefficient at a given half-angle. This coefficient is determined based on its value at normal incidence and the corresponding specular reflection coefficient (given by the Fresnel term computed in Eq. (13)). This formulation ensures compliance with the principle of energy conservation for light interacting with a surface at any incident angle, such that $k_a + k_s + k_d = 1$, where k_a denotes the material's absorption coefficient.

2.2.3. Observed irradiance and apparent magnitude

The irradiance incident on the sensor can be calculated by treating each object facet as a source of reflected sunlight. A small solid angle is assumed, as the characteristic length of the space object is negligible compared to its

distance from the ground-based sensor. Accordingly, the irradiance contribution from each facet γ is:

$$I_{o,\gamma} = \int_{\Omega_{o,\gamma}} (f_{r,s} + f_{r,d})_{\gamma} I_i(\mathbf{v} \cdot \mathbf{n})_{\gamma} d\Omega_{o,\gamma} \approx (f_{r,s} + f_{r,d})_{\gamma} I_i(\mathbf{v} \cdot \mathbf{n})_{\gamma} \frac{A_{\gamma}}{r^2} \quad (17)$$

where A_{γ} is the area of the facet and r is the distance from the space object to the sensor. The total observed irradiance I_o is then computed as the sum of the individual contributions from all facets:

$$I_o = \sum_{\gamma} I_{o,\gamma} \quad (18)$$

The intensity of light reflected by the space object and perceived by the sensor can be better described in terms of apparent magnitude. Since the object's total irradiance is considered (i.e. the sum of the contributions from all wavelengths of the electromagnetic spectrum), the apparent bolometric magnitude is used:

$$m = -2.5 \log \left(\frac{I_o}{I_{\text{ref}}} \right) + T(\lambda, h, \theta, aod_{550}) \quad (19)$$

where $I_{\text{ref}} = 2.518 \times 10^{-8} \text{ W/m}^2$, which corresponds to the zero point of the apparent bolometric magnitude scale.

An atmospheric transmission factor $T(\lambda, h, \theta, aod_{550})$ has been included in Eq. (19) to consider the light extinction caused by Earth's atmosphere. The atmospheric extinction of light depends on Rayleigh scattering (scattering by air molecules), ozone absorption, and diffusion by aerosols suspended in the atmosphere, which depends on the Aerosol Optical Depth (AOD). The model described in Buil (2009) has been used to account for all these contributions to light extinction. This model considers factors like the mean wavelength of light λ , the ground station's altitude above mean sea level h , the elevation angle of the observed space object relative to the ground station θ , and the optical depth of atmospheric aerosols at a wavelength of 550 nm aod_{550} .

2.3. Adaptive importance sampling

In this section, Adaptive Importance Sampling (AIS) is explained in detail, as it is the core estimation technique integrated into the AISwarm-UKF light curve inversion method. The explanation follows a structured approach, starting from general principles and gradually moving to specific details, to justify the selection of AIS and provide a solid understanding of its foundations. The work of Bugallo et al. (2017) provides a comprehensive description of AIS and other related methods, and most of the explanations and derivations presented below are adapted from it.

Statistical inference is a technique used to estimate unknown parameters of a probability distribution based on observed data. Let us consider a generic inference problem in which a d_x -dimensional vector of unknown static

real parameters $\mathbf{x} \in X \subseteq \mathbb{R}^{d_x}$ has a probability density function (pdf) given by:

$$\tilde{\pi}(\mathbf{x}) = \frac{\pi(\mathbf{x})}{Z}, \quad Z = \int_X \pi(\mathbf{x}) d\mathbf{x} \quad (20)$$

where $\pi(\mathbf{x})$ is a non-normalised non-negative target function and Z is a finite normalising constant that may be unknown. The objective is to compute a statistical moment (mean, variance or higher-order moments) of $\mathbf{x}$ or, more generally, the expectation of a function $f(\mathbf{x})$ with respect to the distribution $\tilde{\pi}(\mathbf{x})$:

$$\mathbb{E}_{\tilde{\pi}}[f(\mathbf{x})] = \int_X f(\mathbf{x}) \tilde{\pi}(\mathbf{x}) d\mathbf{x} \quad (21)$$

where $f(\cdot)$ can be any function of $\mathbf{x}$ that is integrable with respect to $\tilde{\pi}(\mathbf{x})$.

The light curve inversion problem can be posed as a Bayesian inference problem to estimate the unknown state parameters from the experimental observations. Bayesian inference resorts to Bayes' theorem to compute the probability of a hypothesis on the unknowns by integrating the observed data with prior knowledge, thus generating a posterior probability distribution that can be used to make inference about the unknowns. Bayes' theorem can be stated mathematically as:

$$\tilde{\pi}(\mathbf{x}) = p(\mathbf{x}|\mathbf{z}) = \frac{\ell(\mathbf{z}|\mathbf{x})p_0(\mathbf{x})}{Z(\mathbf{z})} \propto \ell(\mathbf{z}|\mathbf{x})p_0(\mathbf{x}) = \pi(\mathbf{x}) \quad (22)$$

where $p(\mathbf{x}|\mathbf{z})$ is the posterior pdf linked to some observed data $\mathbf{z}$, $\ell(\mathbf{z}|\mathbf{x})$ is the likelihood function, $p_0(\mathbf{x})$ is the prior pdf, and $Z(\mathbf{z})$ is the model evidence or partition function. $Z(\mathbf{z})$ is generally unknown and difficult to obtain, so hereafter is considered as a normalising constant Z of the non-normalised target density function $\pi(\mathbf{x})$.

In most practical applications it is not possible to analytically compute the integral in Eq. (21). Monte Carlo (MC) methods are based on drawing N independent samples $\{\mathbf{x}_i\}_{i=1}^N$ from the target distribution $\tilde{\pi}(\mathbf{x})$ to approximate the integral with the Monte Carlo estimator:

$$\mathbb{E}_{\tilde{\pi}}[f(\mathbf{x})] \approx \frac{1}{N} \sum_{i=1}^N f(\mathbf{x}_i), \quad \mathbf{x}_i \sim \tilde{\pi}(\mathbf{x}) \quad (23)$$

which is a consistent estimator of expectation because it converges almost surely to $\mathbb{E}_{\tilde{\pi}}[f(\mathbf{x})]$ by the strong law of large numbers. With the drawn samples, the target probability distribution corresponding to the density $\tilde{\pi}(\mathbf{x})$ can be approximated as:

$$\tilde{\pi}^N(\mathbf{x}) \approx \frac{1}{N} \sum_{i=1}^N \delta(\mathbf{x} - \mathbf{x}_i) \quad (24)$$

where $\delta(\mathbf{x} - \mathbf{x}_i)$ is the Dirac delta function centred at $\mathbf{x}_i$.

A widely adopted family of MC methods are the Markov Chain MC (MCMC) algorithms, which generate samples by constructing a Markov chain whose equilibrium distribution matches the target probability distribution of interest. However, MC methods typically assume that

independent samples can be drawn directly from the target distribution, but in many practical scenarios like the one under analysis in this work, this is not possible because the posterior distribution is not known or can only be evaluated up to a normalising constant.

Importance Sampling (IS) methods represent a significant alternative to MCMC algorithms that do not require a priori knowledge of the posterior distribution. The theoretical foundations of IS methods are explained below considering the case where the target distribution given in Eq. (20) can only be known up to a normalising constant.

Let $q(\mathbf{x})$ be a pdf of a distribution with heavier tails than the target pdf from which N independent samples $\{\mathbf{x}_i\}_{i=1}^N$ can be easily drawn. This pdf is called the proposal or the importance density. Then, the integral in Eq. (21) can be transformed as follows assuming that $q(\mathbf{x}) > 0$ whenever $\tilde{\pi}(\mathbf{x})f(\mathbf{x}) \neq 0$:

$$\begin{aligned}\mathbb{E}_{\tilde{\pi}}[f(\mathbf{x})] &= \frac{1}{Z} \int_{\mathcal{X}} q(\mathbf{x}) \left[\frac{\pi(\mathbf{x})}{q(\mathbf{x})} f(\mathbf{x}) \right] d\mathbf{x} \\ &= \frac{1}{Z} \int_{\mathcal{X}} q(\mathbf{x}) [w(\mathbf{x})f(\mathbf{x})] d\mathbf{x} \\ &= \frac{1}{Z} \mathbb{E}_q[w(\mathbf{x})f(\mathbf{x})]\end{aligned}\quad (25)$$

where a new weighting function $w(\mathbf{x})$ has been introduced. In the context of Bayesian inference, this weighting function can be formulated in terms of the posterior pdf and the prior pdf by considering the non-normalised target distribution $\pi(\mathbf{x})$ as defined in Eq. (22):

$$w(\mathbf{x}) = \frac{\pi(\mathbf{x})}{q(\mathbf{x})} = \frac{\ell(\mathbf{z}|\mathbf{x})p_0(\mathbf{x})}{q(\mathbf{x})}\quad (26)$$

The Monte Carlo estimator of the expected value of $w(\mathbf{x})f(\mathbf{x})$ with respect to the importance density $q(\mathbf{x})$ is:

$$\mathbb{E}_q[w(\mathbf{x})f(\mathbf{x})] \approx \frac{1}{N} \sum_{i=1}^N w(\mathbf{x}_i)f(\mathbf{x}_i), \quad \mathbf{x}_i \sim q(\mathbf{x})\quad (27)$$

where $w(\mathbf{x}_i)$ are the importance weights associated with each sample of the proposal density $q(\mathbf{x})$ and quantify their relevance in approximating the target distribution $\tilde{\pi}(\mathbf{x})$. An estimator of the normalising constant can be obtained as:

$$Z = \int_{\mathcal{X}} \left[\frac{\pi(\mathbf{x})}{q(\mathbf{x})} \right] q(\mathbf{x}) d\mathbf{x} \approx \frac{1}{N} \sum_{i=1}^N w(\mathbf{x}_i), \quad \mathbf{x}_i \sim q(\mathbf{x})\quad (28)$$

which can be substituted in Eq. (25) to obtain the importance sampling approximation:

$$\mathbb{E}_{\tilde{\pi}}[f(\mathbf{x})] \approx \sum_{i=1}^N \bar{w}(\mathbf{x}_i)f(\mathbf{x}_i), \quad \mathbf{x}_i \sim q(\mathbf{x})\quad (29)$$

where $\bar{w}(\mathbf{x}_i)$ are the normalised weights of the samples, obtained by:

$$\bar{w}(\mathbf{x}_i) = \frac{w(\mathbf{x}_i)}{\sum_{i=1}^N w(\mathbf{x}_i)}\quad (30)$$

Finally, the target probability distribution can be readily approximated as:

$$\tilde{\pi}^N(\mathbf{x}) \approx \sum_{i=1}^N \bar{w}(\mathbf{x}_i) \delta(\mathbf{x} - \mathbf{x}_i)\quad (31)$$

The main challenge in IS methods lies in selecting proposal densities that facilitate efficient and effective sampling. Adaptive Importance Sampling (AIS) methods overcome this by iteratively refining a proposal or set of proposal densities to better match the target distribution. This refinement is guided by information obtained from previous samples, ultimately improving both the accuracy and reliability of the inference process.

AIS has been selected to address the light curve inversion problem in this work due to several advantages over other MC methods, particularly MCMC methods (Elvira et al., 2017):

- In MCMC, early samples often fail to accurately represent the target distribution because the Markov chain has not yet reached stationarity. These initial samples must be discarded during a burn-in phase. In contrast, AIS incorporates all samples into the estimation process from the outset, regardless of how well the initial proposal distribution aligns with the target distribution.
- AIS methods offer greater flexibility in their adaptive schemes, as convergence is typically ensured under relatively mild conditions —such as the proposal distribution sufficiently covering the support of the target distribution and the adaptive process remaining stable. Conversely, in MCMC, adapting the proposal distribution directly influences the behaviour of the Markov chain. If not carefully designed, this adaptation can compromise the ergodicity of the chain, a fundamental property that guarantees thorough exploration of the target distribution.

AIS consists of three fundamental steps: sampling from the proposal densities, computing the importance weights of each sample, and updating the proposal densities for the subsequent iteration. While a comprehensive review of modern AIS methods can be found in Bugallo et al. (2017), this work introduces a novel adaptive methodology, AISwarm-UKF, specifically designed to address the challenges of the light curve inversion problem for attitude estimation in an operational context.

The AISwarm-UKF method leverages the combined effect of Systematic Resampling and Particle Swarm Optimisation to improve convergence efficiency without compromising estimation accuracy. The theoretical foundations of these techniques are presented in Sections 2.4 and 2.5, respectively, while Section 2.8 details their integration within the overall AIS algorithm.

2.4. Systematic resampling

Resampling is a statistical technique that involves repeatedly drawing samples from a dataset to improve the accuracy of the probability distribution estimate. It is

frequently used in sequential Monte Carlo methods, such as particle filters, to address the particle degeneracy problem (Daum and Huang, 2011). This problem occurs when a small number of particles have significant weight, while the majority contribute little to the overall estimate. In such cases, computational resources are inefficiently used on particles that provide minimal or no valuable contribution. The effective sample size, which provides a measure of particle degeneracy, is given by the following expression (Arulampalam et al., 2002):

$$N_{eff} = \frac{1}{\sum_{i=1}^N \bar{w}_i^2} \quad (32)$$

where $\bar{w}_i$ are the normalised weights of each particle and N is the total number of particles.

The resampling approach used in this work selects new particles with replacement from the set of weighted particles. As a result, the total number of particles remains constant. Particles with higher normalised weights are more likely to be selected multiple times, while particles with lower normalised weights tend to be discarded.

Among the different resampling algorithms available in the literature, Systematic Resampling (Kitagawa, 1996) has been selected for the AISwarm-UKF attitude estimation method because it is simple to implement, takes $\mathcal{O}(N)$ time, and minimises the MC variation (Arulampalam et al., 2002).

The key idea of Systematic Resampling is to select particles at regular intervals across the cumulative distribution function given by the particles weight by dividing the whole population of particles into subpopulations. To that end, the $(0, 1]$ interval is divided into N disjoint subintervals, which represent the portions of the cumulative distribution function where each particle will be selected:

$$(0, 1] = \bigcup_{j=1}^N \left(\frac{j-1}{N}, \frac{j}{N} \right] \quad (33)$$

The cumulative weight for the i -th particle is the cumulative sum of the normalised weights:

$$C_i = \sum_{j=1}^i \bar{w}_j \quad (34)$$

Then, random numbers are generated according to Eq. (35) to obtain stochastic indices that will be mapped to the particles, helping select particles in proportion to their importance weights and ensuring that particles with higher weights are more likely to be selected.

$$\begin{aligned} u_1 &\sim U\left(0, \frac{1}{N}\right] \\ u_k &= u_1 + \frac{k-1}{N}, \quad k = 2, 3, \dots, N \end{aligned} \quad (35)$$

As it can be noticed, only the first number is generated randomly, while the subsequent ones are obtained deterministically. This approach ensures that the resampling process

begins from a random position within the first subinterval, avoiding a deterministic particle selection method.

Once the random numbers are obtained, they are mapped to particle indices by determining which subinterval each random number falls into. Specifically, the particle index i is the smallest index such that the cumulative weight C_i exceeds u_k .

2.5. Particle Swarm Optimisation

Particle Swarm Optimisation (PSO) (Kennedy and Eberhart, 1995) is a population-based global optimisation algorithm inspired by the collective behaviours observed in bird flocking and fish schooling. PSO has been successfully applied in a wide range of application domains (Gad, 2022). In this work, PSO is adopted not only for its proven performance but also for its simplicity (its particle update mechanism requires tuning only a few parameters) and for its adaptability, which makes it well suited for integration into the AISwarm method.

In PSO, a group of candidate solutions (particles) move through the search space according to their momentum, which adjusts the trajectory of the particles and accelerates their convergence towards optimal solutions. The momentum of the particles is updated iteratively based on both their own experience and that of the neighbouring particles (the swarm). Let the j -th component of the i -th particle's position and velocity vectors at iteration k be denoted as $x_{i,j}^k$ and $v_{i,j}^k$, respectively. Then, each scalar state parameter and its velocity are updated according to the following equations (Shi and Eberhart, 1998):

$$\begin{aligned} v_{i,j}^k &= w v_{i,j}^{k-1} + c_1 r_1 (x_{i,j}^{pbest} - x_{i,j}^k) + c_2 r_2 (x_j^{gbest} - x_{i,j}^k) \\ x_{i,j}^k &= x_{i,j}^{k-1} + v_{i,j}^k \end{aligned} \quad (36)$$

where w is the inertia weight factor, c_1 is the cognitive coefficient, $x_{i,j}^{pbest}$ is the j -th component of the best state found by the i -th particle (personal best), c_2 is the social coefficient, x_j^{gbest} is the j -th component of the best state found by any particle in the swarm (global best), and r_1 and r_2 are random numbers drawn from $U(0, 1)$.

Proper tuning of the different parameters is problem-dependent and crucial for optimising PSO's performance and accuracy, e.g. avoiding premature convergence to local optima. The inertia weight controls how much of a particle's previous momentum is retained. The cognitive and social coefficients determine how much influence a particle's own experience and the swarm's collective experience, respectively, have on its movement.

A reasonable strategy is proposed in Ratnaweera et al. (2004), where local exploration is promoted in the early stages of the optimisation while, as iterations advance, exploitation is encouraged by accelerating the converge of particles towards global best states. To that end, the cognitive coefficient is decreased linearly and the social coefficient is increased linearly as iterations progress:

$$\begin{aligned} c_1(k) &= c_{1,max} + \frac{k}{K}(c_{1,min} - c_{1,max}) \\ c_2(k) &= c_{2,min} + \frac{k}{K}(c_{2,max} - c_{2,min}) \end{aligned} \quad (37)$$

where K controls the iteration number at which the linear variation of the coefficients is concluded.

2.6. Cluster analysis

Clustering, or cluster analysis, is a technique used to group similar data points based on shared characteristics. It facilitates the identification of patterns within a dataset by organising elements into distinct clusters, where members of the same cluster exhibit greater similarity to one another than to those in different clusters. Clustering is widely applied across various domains to classify data and detect trends, thereby supporting the interpretation of complex information and informing decision-making. There are different types of clustering algorithms, such as hierarchical, partition-based, and density-based methods, each suited to different data types and clustering structures.

The proposed attitude estimation method utilises Density-Based Spatial Clustering of Applications with Noise (DBSCAN) (Ester et al., 1996), a density-based clustering algorithm that identifies clusters according to the density of data points. Unlike partition-based methods such as K-Means, DBSCAN does not require the number of clusters to be specified in advance. Instead, it operates based on two key parameters:

- The neighbourhood radius, which defines the maximum distance within which a point is considered a neighbour of another point.
- The minimum number of points required to form a cluster.

DBSCAN identifies the neighbourhood of each point and designates as core points those that have at least the specified minimum number of neighbours. Core points are then grouped into clusters by connecting directly or indirectly reachable core points. Non-core points are assigned to a nearby cluster if they fall within the neighbourhood radius of a core point. Any point that is not reachable from a core point is classified as an outlier. The DBSCAN implementation from the Hipparchus library (Hipparchus, 2024) has been used in this work.

2.7. Unscented Kalman Filter

The Unscented Kalman Filter (UKF) (Julier and Uhlmann, 1997) is a recursive algorithm designed to estimate the state of non-linear systems. In this work, the UKF is employed to refine the attitude estimate provided by the AISwarm method. It is particularly well-suited to this task because it does not require partial derivatives with respect to the attitude parameters (derivatives that would otherwise need to be computed numerically and are often highly unstable). Additionally, the UKF has been shown

in the literature to provide reliable accuracy when the initial deviation from the true attitude state is relatively small (Wetterer and Jah, 2009).

The UKF accounts for non-linearities by implementing the unscented transformation, which uses a minimal set of carefully chosen sample points called sigma points to approximate the state probability distribution. These sigma points are propagated through the non-linear system dynamics to estimate the predicted state. The predicted measurements are then obtained by applying the non-linear measurement model to the propagated sigma points, providing a more accurate estimation of the state and its covariance compared to methods that rely on linear approximations like the Extended Kalman Filter (EKF).

The state evolution of a non-linear system is governed by the following discrete-time dynamic and measurement models:

$$\begin{aligned} \mathbf{x}_{k+1} &= f(\mathbf{x}_k, \mathbf{v}_k) \\ \mathbf{y}_k &= h(\mathbf{x}_k, \mathbf{w}_k) \end{aligned} \quad (38)$$

where $\mathbf{x}_k$ is the state of the system at time k , $f(\cdot)$ is the non-linear function governing the kinematics described in Section 2.1, $\mathbf{v}_k$ is the process noise with covariance matrix $\mathbf{Q}$, $\mathbf{y}_k$ is the observed measurement signal at time k , $h(\cdot)$ is the non-linear measurement function introduced in Section 2.2, and $\mathbf{w}_k$ is the measurement noise with covariance matrix $\mathbf{R}$.

The sigma points used to approximate the non-linear transformations can be obtained from the available state estimate $\hat{\mathbf{x}}_0$ or $\hat{\mathbf{x}}_{k-1}$ and its associated covariance $\mathbf{P}_0$ or $\mathbf{P}_{k-1}$. In order to estimate the first two moments of the state, $2d_x + 1$ sigma points are generated deterministically:

$$\begin{aligned} \mathcal{X}_{0,k-1} &= \hat{\mathbf{x}}_{k-1} \\ \mathcal{X}_{i,k-1} &= \hat{\mathbf{x}}_{k-1} + \sqrt{d_x + \kappa} \mathbf{A}_i, \quad i = 1, \dots, d_x \end{aligned} \quad (39)$$

$$\mathcal{X}_{i,k-1} = \hat{\mathbf{x}}_{k-1} - \sqrt{d_x + \kappa} \mathbf{A}_i, \quad i = d_x + 1, \dots, 2d_x$$

with the following associated weights:

$$\begin{aligned} W_0 &= \frac{\kappa}{d_x + \kappa} \\ W_i &= \frac{1}{2(d_x + \kappa)}, \quad i = 1, \dots, 2d_x \end{aligned} \quad (40)$$

where d_x is the dimensionality of the state, κ is a scaling parameter, and $\mathbf{A}_i$ is the i -th row of $\mathbf{A}$, with $\mathbf{P}_{k-1} = \mathbf{A}^T \mathbf{A}$. The matrix $\mathbf{A}$ can be obtained with the Cholesky decomposition of $\mathbf{P}_{k-1}$.

The state and the associated covariance at the current time k are obtained in the prediction step. The sigma points are propagated according to the non-linear dynamic model and are weighted to obtain the predicted mean $\hat{\mathbf{x}}_k^-$ and covariance $\mathbf{P}_{x_k}^-$:

$$\mathcal{X}_{i,k} = f(\mathcal{X}_{i,k-1}) \quad (41)$$

$$\hat{\mathbf{x}}_k^- = \sum_{i=0}^{2d_x} W_i \mathcal{X}_{i,k} \quad (42)$$

$$\mathbf{P}_{x_k}^- = \sum_{i=0}^{2d_x} W_i (\mathcal{X}_{i,k} - \hat{\mathbf{x}}_k^-) (\mathcal{X}_{i,k} - \hat{\mathbf{x}}_k^-)^T + \mathbf{Q} \quad (43)$$

The predicted state and measurements are combined with actual measurements to update the state estimate and covariance. The predicted measurements are obtained by applying the measurement model to the predicted sigma points $\mathcal{X}_{i,k}$. Then, the predicted measurement mean $\hat{\mathbf{y}}_k^-$ and covariance $\mathbf{P}_{y_k}$ are calculated similarly to the prediction step according to the weighted sum of the predicted measurements:

$$\mathcal{Y}_{i,k} = h(\mathcal{X}_{i,k}) \quad (44)$$

$$\hat{\mathbf{y}}_k^- = \sum_{i=0}^{2d_x} W_i \mathcal{Y}_{i,k} \quad (45)$$

$$\mathbf{P}_{y_k} = \sum_{i=0}^{2d_x} W_i (\mathcal{Y}_{i,k} - \hat{\mathbf{y}}_k^-) (\mathcal{Y}_{i,k} - \hat{\mathbf{y}}_k^-)^T + \mathbf{R} \quad (46)$$

The next step is to calculate the cross-covariance between the predicted state and the predicted measurement $\mathbf{P}_{x_k y_k}$, necessary to compute the Kalman gain $\mathbf{K}_k$. The Kalman gain determines to what extent the measurement residual needs to be weighted when updating the state estimate. The state estimate $\hat{\mathbf{x}}_k$ is updated by adding the weighted measurement residual, which is computed using the actual measurement $\mathbf{z}_k$ and the predicted measurement mean $\hat{\mathbf{y}}_k^-$. Finally, the updated covariance $\mathbf{P}_{x_k}$ is obtained by adjusting the predicted covariance to reflect the uncertainty after incorporating the new measurement:

$$\mathbf{P}_{x_k y_k} = \sum_{i=0}^{2d_x} W_i (\mathcal{X}_{i,k} - \hat{\mathbf{x}}_k^-) (\mathcal{Y}_{i,k} - \hat{\mathbf{y}}_k^-)^T \quad (47)$$

$$\mathbf{K}_k = \mathbf{P}_{x_k y_k} \mathbf{P}_{y_k}^{-1} \quad (48)$$

$$\hat{\mathbf{x}}_k = \hat{\mathbf{x}}_k^- + \mathbf{K}_k (\mathbf{z}_k - \hat{\mathbf{y}}_k^-) \quad (49)$$

$$\mathbf{P}_{x_k} = \mathbf{P}_{x_k}^- - \mathbf{K}_k \mathbf{P}_{y_k} \mathbf{K}_k^T \quad (50)$$

A dedicated analysis of the light curve inversion problem for attitude estimation has been conducted using the UKF to evaluate its convergence performance. The objective is to determine the threshold for the initial guess error that ensures convergence to the true solution.

To that end, a statistical analysis of the UKF's attitude estimation performance has been conducted using a dataset of 1000 synthetic light curves. All light curves were generated using the same space object model (see Fig. 2), under identical orbital and observation conditions, but with 1000 different sets of orientation parameters. A spin period of 15 s is used in all cases. The sensor noise was modelled with a standard deviation of $\sigma_s = 0.1$ in apparent magnitude. The results of this statistical analysis are shown in Fig. 1.

The box plots show the average Final Orientation Error (FOE) and the Root Mean Square Error (RMSE) of the UKF estimates as a function of the Initial Orientation Error (IOE) and the σ -factor. The average orientation error represents the mean error across all angular state

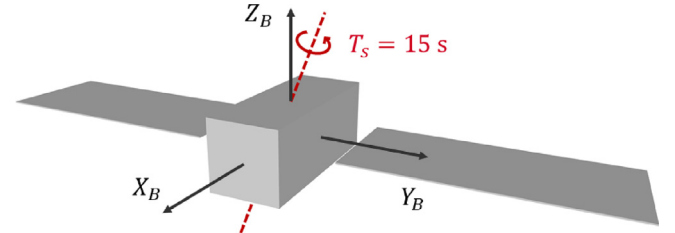


Fig. 2. Satellite model.

parameters, including the object's reference frame orientation (yaw, pitch and roll angles) and the spin axis orientation (azimuth and elevation angles). The σ -factor is a scaling coefficient applied to the IOE to define the initial UKF standard deviation.

The results of this statistical analysis show that the UKF's convergence is highly sensitive to the initial guess, yielding unreliable attitude estimates for average IOEs greater than 5 deg. Moreover, the RMSE remains below 0.4 in most cases, which, for a typical light curve ranging from 4 to 8 in magnitude, corresponds to a RMSE bounded below 10 % of the light curve range. These findings demonstrate the existence of multiple favourable local optima, reinforcing the need for an approach capable of effectively filtering them to identify the true attitude.

2.8. AISwarm-UKF attitude estimation method

The five methods previously described are integrated to address the light curve inversion problem for attitude estimation, offering an accurate, robust, and computationally efficient solution that is suitable for real-world operational scenarios. These methods are complementary, allowing their individual strengths to be maximised while mitigating the limitations of each through the advantages of the others. The proposed methodology is presented in Algorithm 1, along with a detailed explanation.

The AISwarm-UKF attitude estimation method requires several inputs, including the light curves recorded by ground-based optical sensors, the geodetic positions of the observing stations, and the noise characteristics of the sensors (e.g. standard deviation). Additionally, the method needs information about the space object's geometry, optical reflectance properties, orbital parameters, and (inertial) spin period. The orbital information can be provided using a Two-Line Element (TLE) set propagated with the SGP4 model (Hoots and Roehrich, 1980), which reduces computational cost while maintaining sufficient accuracy, as the observation model is highly sensitive to the object's orientation but less so to its exact orbital position. The spin period of the space object could also be estimated using the AISwarm-UKF method. However, since the method is affected by the curse of dimensionality, reducing the number of estimated parameters can significantly improve the computational performance. Therefore, it is more efficient to estimate the spin period beforehand using alternative

approaches, such as the method proposed by Hall et al. (2007).

The AISwarm-UKF method begins by initialising the particle set $\mathcal{P}$. In the absence of prior knowledge about the object's attitude, the state of each particle $\mathbf{x}_i$ is sampled from a uniform distribution that spans the full solution space, respecting the domain constraints of each state parameter. These particles serve as candidate solutions representing the object's attitude at the end of the observation interval.

The next step involves segmenting the M observed light curves $\mathcal{L}_m$ into sub-tracks $\mathcal{L}_{m,p}$, using the inertial rotation period as the sampling interval. Although it is assumed that all sensors collect measurements within roughly the same time frame (ensuring the validity of the constant angular velocity assumption) the number of resulting sub-tracks L_m may vary between light curves. This is not problematic and is handled appropriately during the execution of the AISwarm method. This segmentation allows photometric measurements to be processed in groups at each iteration, thus taking advantage of the previously discussed benefits of a batch-processing strategy.

Algorithm 1. AISwarm-UKF attitude estimation method.

Inputs:

- Observed light curves: $\mathcal{L} = \{\mathcal{L}_m\}_{m=1}^M$
- Ground station locations
- Sensor noise standard deviations: $\sigma_{s,m}$
- Space object's geometry, optical properties, orbital parameters and spin period

Outputs:

- Attitude estimates: $\{(\tilde{\mathbf{x}}_k, \sigma_{\tilde{\mathbf{x}}_k})\}_{k=1}^K$
- RMSE associated with each $\tilde{\mathbf{x}}_k$

```

1: Initialise the particles:
    $\mathcal{P} = \{p_i\}_{i=1}^N = \{(\mathbf{x}_i, \mathbf{v}_i = 0)\}_{i=1}^N$ 
2: for all  $\mathcal{L}_m \in \mathcal{L}$  do
3:    $\mathcal{L}_m = \{\mathcal{L}_{m,p}\}_{p=1}^{L_m} \leftarrow$  Divide  $\mathcal{L}_m$  into sub-tracks
4: end for
5:  $iter \leftarrow 1$ 
6: while  $iter \leq maxIter$  do
7:    $\mathcal{Z} = \{z_j\}_{j=1}^M \leftarrow$  Select measurements from  $\mathcal{L}$ 
8:    $\{\bar{w}_i\}_{i=1}^N \leftarrow ComputeWeights(\mathcal{P}, \mathcal{Z})$ 
9:   if  $iter \leq 2$  then
10:    Perform adaptive particle allocation
11:     $iter \leftarrow iter + 1$ 
12:    continue
13:   end for
14:   if  $iter \geq 5$  then
15:      $\mathcal{C} = \{\mathcal{P}_k\}_{k=1}^K, \mathcal{P}_k = \{p_i\}_{i=1}^{N_k} \leftarrow$  Clustering
16:     if  $\sum N_k / N > R_{\mathcal{C},thr}$  then
17:       for all  $\mathcal{P}_k \in \mathcal{C}$  do
18:         if  $\sigma_{w,k} < \sigma_{thr}$  and  $\Delta_{\mathbf{x}_k} < \Delta_{thr}$  then
19:           Freeze cluster  $\mathcal{P}_k$ 
20:         end if
21:       end for

```

```

22:       if  $\forall \sigma_{w,k} < \sigma_{thr}$  and  $\forall \Delta_{\mathbf{x}_k} < \Delta_{thr}$  then
23:         break
24:       end if
25:     end if
26:     if  $N_{eff} < N_{thr}$  then
27:       Perform Systematic Resampling
28:     end if
29:     Execute the PSO method
30:      $iter \leftarrow iter + 1$ 
31:   end while
32:   for all  $\mathcal{P}_k \in \mathcal{C}$  do
33:      $\tilde{\mathbf{x}}_k, \sigma_{\tilde{\mathbf{x}}_k} \leftarrow$  Execute the UKF
34:     Compute  $RMSE_k$ 
35:   end for
36:   function ComputeWeights $\mathcal{P}_{in}, \mathcal{Z}_{in}$ 
37:     for all  $p_i \in \mathcal{P}_{in}$  do
38:       for all  $z_j \in \mathcal{Z}_{in}$  do
39:         Propagate  $\mathbf{x}_i$  to the time of  $z_j$ 
40:         Simulate the photometric measurement  $y_j$ 
41:       end for
42:       Compute  $WRMSE_i$  and  $w_i$ 
43:     end for
44:     Normalise weights
45:   return  $\{\bar{w}_i\}_{i=1}^{N_{in}}$ 
46: end function

```

Once the particles have been initialised and the observed light curves have been divided into sub-tracks, the execution of the AISwarm method can commence. At each iteration, it is necessary to select the real measurements $\mathcal{Z}$ to be used, which is achieved by choosing sub-tracks from the previously segmented light curves. This selection follows the criteria outlined below:

- An attempt is made to include at most one sub-track from each light curve in every iteration.
- Within each light curve, the selection of the sub-track is performed randomly.
- Priority is given to sub-tracks that have not yet been processed, ensuring that all sub-tracks across all light curves are utilised before any are repeated.

This measurement selection strategy is designed to guarantee an equal contribution from all sub-tracks to the attitude estimation process. Within each light curve, sub-tracks are chosen randomly rather than in temporal order to improve the robustness of the methodology and reduce the risk of convergence to local optimal solutions, particularly when only a single light curve is available.

The next step involves computing the particle quality metrics: the weight and the Weighted Root Mean Square Error (WRMSE). To achieve this, each particle state is first propagated to the time of the selected measurements using the kinematic model described in Section 2.1. Simulated measurements are then generated using the observation

model explained in Section 2.2. These computations are parallelised to improve the computational efficiency.

The WRMSE is selected as particle quality metric due to the batch processing of photometric measurements, which requires a metric that quantifies the deviation between real and simulated sub-tracks originating from multiple sensors with different noise levels. The WRMSE is a more stable metric than the particle weight with respect to variations in the particles' state and is more suitable for establishing relative thresholds. Therefore, it is used to guide the adaptive allocation of particles in the initial iterations of the AISwarm method, ensuring robustness in detecting all candidate solutions. The particle WRMSE is computed as (DiStefano et al., 2018):

$$WRMSE = \sqrt{\frac{1}{M} \sum_{j=1}^M \left(\frac{y_j - z_j}{\sigma_{s,j}} \right)^2} \quad (51)$$

where z_j is a real photometric observation, $y_j = y_j(\mathbf{x}_i, t_{z_j})$ is a simulated measurement obtained for the particle state $\mathbf{x}_i$ at the time of the real measurement z_j , M is the number of measurements of the sub-track, and $\sigma_{s,j}$ denotes the sensor noise standard deviation associated with measurement z_j .

The particle weights are used in the Systematic Resampling and Clustering steps. They are computed using Eq. (26), which can be further simplified if the importance density is selected to be the prior density at each iteration of the AISwarm method, i.e. $q(\mathbf{x}) = p_0(\mathbf{x})$:

$$w(\mathbf{x}_i) = \ell(\{z_j\}_{j=1}^M | \mathbf{x}_i) = \exp(-WRMSE_i^2) \quad (52)$$

where the likelihood function represents the probability of observing a set of photometric measurements given the particle state. As indicated in Eq. (52), the particle WRMSE is mapped to a probability using a (still non-normalised) softmax function. The weights are normalised afterwards according to Eq. (30).

The adaptive allocation of particles is performed during the first two iterations to refine the initial proposal distribution. This process consists of relocating particles from low-probability regions to regions of higher probability. Specifically, particles with a WRMSE within the highest tertile of the WRMSE distribution are repositioned near the top-performing particles. The new positions are computed using the sampling step employed in generating the initial distribution, placing the relocated particles at intermediate positions between the top-performing ones. This strategy increases the concentration of particles in high-probability regions, thereby improving both estimation accuracy and convergence efficiency in subsequent AISwarm iterations.

The cluster analysis is conducted using the DBSCAN method to assess whether the particles have sufficiently converged to high-probability regions. The clustering algorithm is computationally costly and is therefore only executed after four iterations of the AISwarm method. To ensure a robust identification of all clusters, conservative

clustering parameters are selected. The minimum number of particles required to form a cluster is set to 10 % of the total number of particles, and the neighbourhood radius is defined as 20 % of the UKF convergence threshold $\sigma_{w,thr}$, whose determination is explained below together with the local convergence criterion.

The cluster analysis serves two primary objectives: first, to verify whether the majority of particles have converged to high-probability regions, and second, to evaluate whether the identified candidate solutions are sufficiently accurate for refinement using the UKF, a computationally more efficient approach. To ensure these conditions are met, two convergence criteria are applied: one assessing global convergence and another specific to each particle cluster.

- Global criterion: The number of particles not associated with any cluster must be less than 20 % of the total number of particles N , i.e. $R_{\mathcal{C},thr} = 80 \%$.
- Local criterion: The weighted standard deviation within each cluster $\sigma_{w,k}$ must fall below the UKF convergence threshold $\sigma_{w,thr}$, and the change in each state parameter between consecutive iterations $\Delta_{\mathbf{x}_k}$ must be lower than $\Delta_{thr} = 10 \%$.

The local convergence threshold of the UKF is determined based on the statistical analysis presented in Fig. 1. A threshold of $\sigma_{w,thr} = 5$ deg is selected for each state parameter, as this IOE value shows good convergence performance. This threshold is considered conservative, especially given that the space object model used in the analysis is simple and symmetric, which is inherently more challenging due to the higher measurement ambiguities, as discussed earlier.

It is important to specifically address the clustering process when the elevation angle of the spin axis is near ± 90 deg, as the azimuth angle becomes indeterminate in such cases. To ensure robustness in the identification of the clusters, the clustering metric excludes the azimuth angle when the elevation is greater than 89 deg or lower than -89 deg. Once the clusters are identified, the dispersion of particles in azimuth becomes more challenging to reduce as the angle approaches these extreme values. Consequently, the convergence threshold is not applied to the azimuth if all the other state parameters have already converged.

As indicated in Algorithm 1, the local convergence analysis is only performed once the global convergence criterion has been met. If a cluster satisfies the local convergence criterion, it is considered "frozen", meaning that its particles are no longer updated in subsequent iterations. The AISwarm method continues to run until either all clusters meet the local convergence criterion or the user-defined maximum number of iterations is reached.

If the global criterion is not satisfied, or if any cluster fails to meet the local criterion, the associated particles are updated for the next iteration with Systematic Resampling and PSO. Systematic Resampling is first applied as

explained in Section 2.4, with $N_{thr} = N/4$, to prevent particle degeneracy. Subsequently, their states are updated using PSO, which moves the particles towards more optimal regions of the parameter space following the methodology outlined in Section 2.5.

The use of PSO allows the target distribution to be estimated implicitly, avoiding the computational cost of explicitly calculating it at each iteration. As the particles converge towards high-probability regions of the parameter space, their updated states naturally form a refined proposal distribution. This approach is also more efficient than traditional methods used in particle filters to maintain particle diversity, such as injecting artificial noise, ultimately leading to faster and more reliable convergence towards the correct attitude estimates.

In the AISwarm method, the initial exploration of the solution space is primarily driven by the adaptive allocation of particles during the first few iterations. As a result, although PSO inherently supports exploration, it is specifically configured here to focus on accelerating convergence towards optimal solutions. To this end, the social coefficient is consistently set higher than the cognitive coefficient throughout the iterations. The linear variation of the social and cognitive coefficients, defined in Eq. (37), is based on the following parameters, which were determined experimentally: $c_{1,min} = 0.5$, $c_{1,max} = 1.5$, $c_{2,min} = 1.0$, $c_{2,max} = 2.0$, and $K = 3$. Once the number of PSO iterations — which refers to the number of times PSO is executed and should not be confused with the iteration count of the complete AISwarm method — exceeds K , the values of c_1 and c_2 remain fixed at their values from iteration K . The inertia weight factor is kept constant across all iterations, with a value of $w = 0.3$.

Since the target probability distribution is likely to be multimodal, the PSO maintains a set of global best states, which is updated at each iteration based on two criteria that must be simultaneously satisfied:

- Selected particles must have a WRMSE within the lowest tertile of the current iteration's WRMSE distribution. Replicated particles resulting from the resampling step, if any, are considered only once. Global best states from the previous iteration that meet this WRMSE threshold are also selected.
- Only the particle with the lowest WRMSE within a defined neighbourhood is retained. Neighbourhoods are defined based on the assumption that different attitude solutions do not exist within an angular separation of 30 deg. Two particles are considered to belong to the same neighbourhood if the angular distance between their respective main body orientations, spin axis directions, or the combined angular distance, is lower than the specified angular threshold.

At each iteration of the PSO, the velocity of each particle is updated considering its nearest global best state. To ensure gradual convergence towards the most promising

attitude estimates, the nearest global best state of each particle is identified using the Manhattan distance (Han et al., 2022), computed across all state parameters while appropriately accounting for the periodic nature of some of them.

Once convergence is achieved, the candidate solutions identified in the cluster analysis are refined using the UKF with all available photometric measurements. The most probable (weighted) attitude state within each cluster serves as the initial guess for the UKF. The initial standard deviation for each state parameter is set to 5 deg, as this IOE has demonstrated good convergence performance in the statistical analysis presented in Fig. 1. However, if the weighted standard deviation of the spin axis azimuth angle has not converged below 5 deg, its actual value is used as the initial guess instead.

Finally, the Root Mean Square Error (RMSE) associated with each refined attitude solution is computed using the available photometric measurements, serving as a quantitative metric for comparing different solutions and assessing their reliability.

3. Results

This section presents the application of the proposed AISwarm-UKF methodology for estimating the attitude of an uncontrolled space object. The case study is representative of a realistic operational scenario, for example a situation in which a satellite operator has lost control of a satellite and must estimate its rotational state to assess the feasibility of In-Orbit Servicing (IOS) or Active Debris Removal (ADR) operations.

The satellite is assumed to be in a circular orbit at an altitude of 2000 km, within the Low Earth Orbit (LEO) region, with an orbital inclination of 10 deg and a right ascension of the ascending node of 0 deg. The orbital altitude is intentionally set higher than typical for LEO objects to facilitate the simulation of stereoscopic observations from two ground stations within the same observation interval and with notably different Sun-object-sensor geometries. This assumption does not constitute a limitation for practical applications, as the proposed method is capable of accurately estimating the attitude state using a single light curve from a single sensor, as demonstrated in this section.

The satellite model is depicted in Fig. 2, showing that it consists of a platform —modelled as a quadrangular prism with dimensions $2.5 \times 1 \times 1$ m— and a solar array with dimensions $1.5 \times 3.5 \times 0.01$ m. The total span of the satellite is 9 m. The optical properties of the surfaces are assumed to be known: the platform is characterised by a diffuse reflection coefficient of $k_d = 0.15$ (Mulrooney et al., 2008), while the solar arrays exhibit purely specular behaviour with $k_s = 0.1$ (Hall, 2010). All surfaces are modelled with a roughness coefficient of $n_s = 0.1$. The inertial rotation period is also assumed to be known and is set to

15 s, since it can be estimated using alternative, more computationally efficient methods, as discussed in Section 2.8.

This case study is based on simulated photometric measurements that closely replicate the light curve that would be observed by a real ground-based optical sensor. The reference attitude used to generate the light curve is defined by the yaw ψ , pitch θ and roll ϕ angles, which specify the orientation of the body-fixed reference frame with respect to the inertial GCRF, and the azimuth α and elevation ϵ angles, which describe the orientation of the spin axis relative to the body-fixed frame:

$$\mathbf{x}_{ref} = \{\psi, \theta, \phi, \alpha, \epsilon\}^T = \{18, 37, -12.5, 110, 81\}^T \text{ deg}$$

The satellite model in this reference attitude produces the light curve shown in Fig. 3, as observed from a hypothetical ground station located at 0°N, 0°E, 1300 m above mean sea level (WGS84). The light curve is simulated with a sensor noise standard deviation of $\sigma_s = 0.1$ in apparent magnitude. The atmospheric conditions at the ground station are assumed to be known, with the Aerosol Optical Depth set to $aod_{550} = 0.15$.

For a space object in a 2000 km altitude orbit, the visibility interval from a ground station can exceed 15 min. However, in this study, only a portion of the light curve is considered. This selection is motivated by the subsequent analysis involving stereoscopic observations, which requires the observation intervals from both ground stations to be approximately simultaneous to ensure the validity of the constant angular velocity assumption. Accordingly, the same time interval is selected for both sensors, chosen to cover the period with the highest irradiance levels at both stations.

The execution of the AISwarm-UKF method is described below. Fig. 4 illustrates the distribution of particles across the different iterations of the AISwarm method, with the actual attitude state to be estimated marked by a red cross. Firstly, the light curve is divided into 16 sub-

tracks, corresponding to the number of complete satellite rotations during the selected observation interval, based on the known inertial rotation period. A uniform distribution of particles is then generated across the solution space, constrained by the symmetric geometric and optical properties of the satellite model. The sampling resolution was determined experimentally through the analysis of multiple scenarios. A step size of 32 deg for each angular state parameter was found to provide a sufficiently fine discretisation to solve the light curve inversion problem both accurately and robustly. This configuration results in a total of 14256 particles.

Due to the constraints imposed on the solution space by the satellite's symmetric geometric and optical properties, any solutions identified within the defined search domain will have corresponding symmetric attitude states. Discriminating between these symmetric solutions to determine a unique attitude estimate is inherently impossible, even when stereoscopic measurements are employed. This ambiguity can only be resolved if the symmetry in either the geometric or optical properties of the space object model is broken.

The first two iterations of the AISwarm method are dedicated to exploring the search space through adaptive particle allocation, ensuring the identification of all promising candidate solutions. As shown in Fig. 4, by the third iteration, the regions of highest probability density have already been identified. In the subsequent iterations, the combined effects of systematic resampling and PSO enable a more focused exploration of these high-probability regions, thereby accelerating convergence towards the most likely attitude states. By the fifth iteration, two principal modes of the posterior probability density function are clearly distinguished. At this stage, 83.24 % of the particles are contained within a cluster, satisfying the global convergence criterion. The cluster associated with Mode 1 satisfies the local convergence criterion in the sixth iteration, whereas

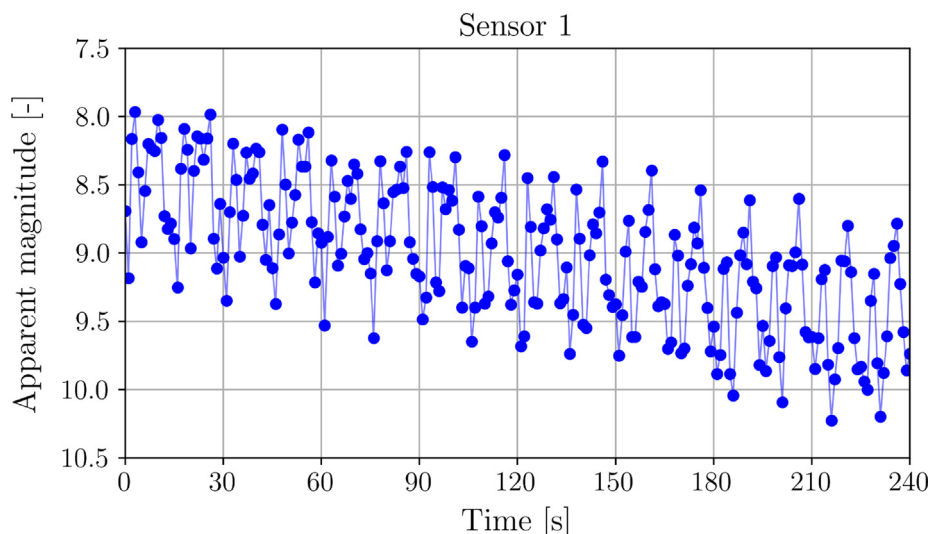
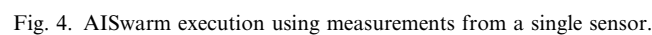


Fig. 3. Light curve of Sensor 1.



Mode 2 requires one additional iteration to converge. In addition to the two dominant modes, two smaller particle groups persist from the fifth iteration onwards. However, these groups do not meet the minimum particle count required to form a valid cluster and correspond to local optima. In fact, these particles would eventually converge to one of the two dominant modes if the AISwarm method were executed through all the sub-tracks, that is, up to the sixteenth iteration in Fig. 4.

Table 1 summarises the results obtained by the AISwarm method upon achieving local convergence of the clusters, that is, prior to refining the solutions using the UKF. The table also presents the final attitude estimates provided by the complete AISwarm-UKF method for the two modes of the posterior probability density function that have been identified. The reference state of Mode 1 is the attitude used to generate the light curve x_{ref} , while the reference state of Mode 2 corresponds to the AISwarm-UKF estimate obtained after allowing the AISwarm method to proceed until the last iteration, thereby incorporating all the available sub-tracks into the estimation process. The final standard deviation of the AISwarm-UKF attitude estimates is also reported, reflecting the confidence level associated with each solution.

The light curves corresponding to the final AISwarm-UKF attitude estimates are shown in Fig. 5. To further evaluate the accuracy of the solutions, the residuals between each simulated measurement and the corresponding reference light curve values are computed and normalised by the standard deviation of the sensor noise. The resulting normalised residuals for both candidate solutions are shown in Fig. 6. For Mode 1, 97.10 % of the photometric measurements fall within the ± 2 normalised residual range, while for Mode 2, the corresponding percentage is 82.99 %.

Finally, the Root Mean Square Error (RMSE) of both solutions relative to the reference light curve is computed. Mode 1 yields a RMSE of 0.0988, whereas Mode 2 results in a RMSE of 0.1390. The RMSE of Mode 1 deviates by only 1.2 % from the standard deviation of the sensor noise used to generate the reference light curve ($\sigma_s = 0.1$), while that of Mode 2 deviates by 39 %. These results, together with the normalised residuals shown in Fig. 6, indicate that Mode 1 offers the best attitude estimate and, in fact, corresponds to the reference attitude state.

The discrepancy between the two modes suggests the presence of an imperfect measurement ambiguity, which, in certain cases, may result in quantitatively similar light curves that are difficult to distinguish. The subsequent analysis explores the potential of stereoscopic measurements to address this ambiguity more directly, without the need for a posteriori residual analysis that may depend on subjective thresholds to differentiate between candidate solutions.

Stereoscopic photometric measurements are acquired by optical sensors located at different ground stations, observing the satellite over approximately the same time interval. In addition to the light curve shown in Fig. 3 for Sensor 1, the light curve from Sensor 2, shown in Fig. 7, is also considered. This second light curve is simulated for a hypothetical sensor located at 15°N, 25°E, 1800 m above mean sea level (WGS84). The sensor noise standard deviation and the Aerosol Optical Depth are again set to $\sigma_s = 0.1$ and $aod_{550} = 0.15$, respectively.

In this test case, the use of stereoscopic measurements is expected to reduce the number of local optima. Consequently, the sampling step for the initial particle distribution can be increased to 35 deg, resulting in a total of 6250 particles. Although the number of available photometric measurements is doubled compared to the previous case, approximately the same number is used in each iteration to ensure a fair comparison with the first test case. This is achieved by randomly selecting measurements from the corresponding sub-tracks of both light curves.

Fig. 10 shows the evolution of the particle distribution across successive iterations of the AISwarm method. In this test case, the posterior probability density function exhibits a single dominant mode, effectively resolving the measurement ambiguity observed in the first test case. The global convergence threshold is reached by the fifth iteration. However, since the number of particles is significantly lower than in the single light curve scenario, the cluster requires until the eighth iteration to meet the local convergence threshold and provide a reliable initial estimate for the UKF.

Table 2 presents the AISwarm and the AISwarm-UKF estimates, along with the final standard deviation of each attitude parameter. The comparison with the reference attitude state confirms the high accuracy of the estimation method. This is further illustrated in Fig. 8, where the sim-

Table 1
Results of the analysis with measurements from a single sensor.

	Mode 1			Mode 2		
	Reference	AISwarm	AISwarm-UKF	Reference	AISwarm	AISwarm-UKF
Yaw [deg]	18	16.65	17.68 $\pm$ 0.33	24.47	24.28	24.39 $\pm$ 0.32
Pitch [deg]	37	40.79	36.97 $\pm$ 0.18	−63.23	−57.66	−61.74 $\pm$ 0.22
Roll [deg]	−12.5	−14.41	−12.80 $\pm$ 0.20	−36.17	−35.69	−36.12 $\pm$ 0.26
Azimuth [deg]	110	102.46	109.97 $\pm$ 0.70	280.0	281.47	284.31 $\pm$ 0.96
Elevation [deg]	81	79.80	80.82 $\pm$ 0.13	82.21	82.78	81.89 $\pm$ 0.14

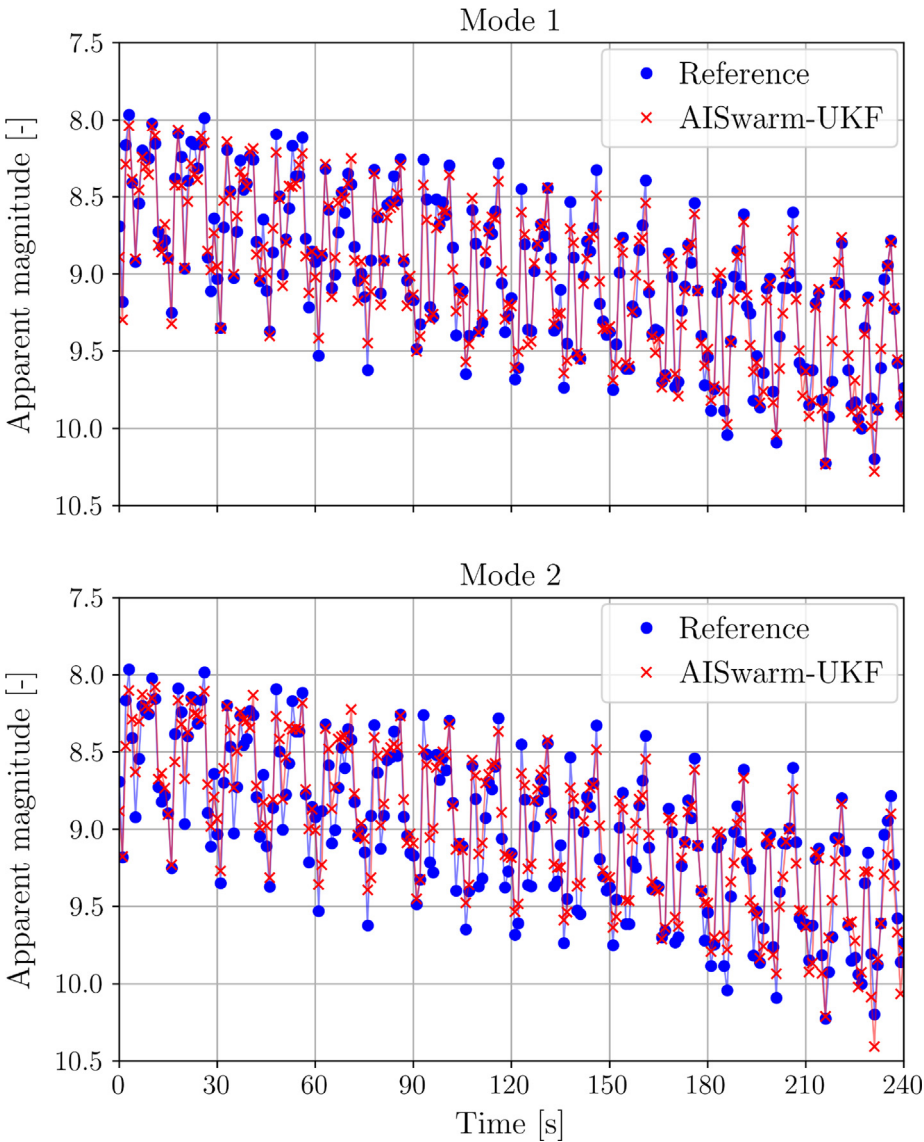


Fig. 5. Comparison of estimated and reference light curves.

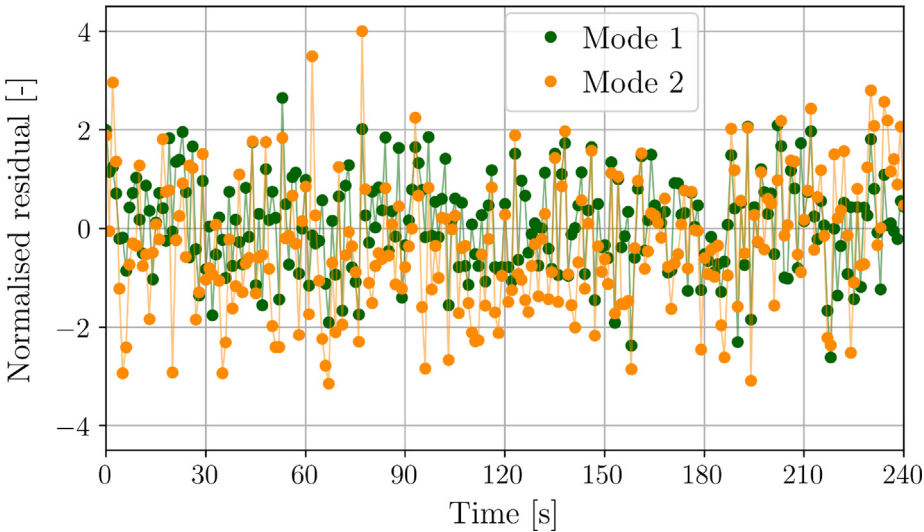


Fig. 6. Relative error between the estimated and reference light curves.

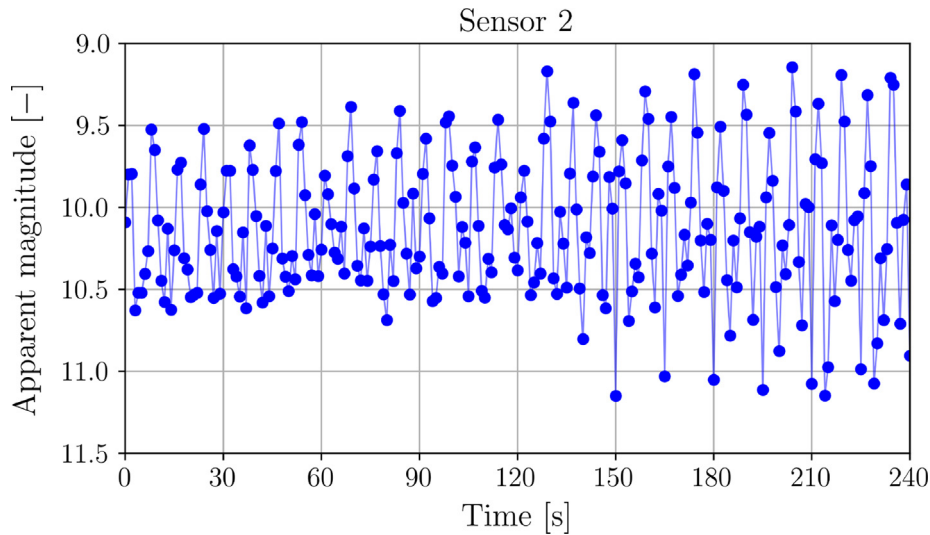


Fig. 7. Light curve of Sensor 2.

Table 2
Results of the analysis with stereoscopic measurements.

	Reference	AISwarm	AISwarm-UKF
Yaw [deg]	18	16.82	18.04 ± 0.35
Pitch [deg]	37	35.21	36.65 ± 0.18
Roll [deg]	-12.5	-13.33	-12.53 ± 0.20
Azimuth [deg]	110	102.71	112.25 ± 0.70
Elevation [deg]	81	80.66	80.72 ± 0.14

ulated light curves corresponding to the estimated attitude closely match the reference ones. The RMSE relative to the reference light curves is 0.0985 for Sensor 1 and 0.0975 for Sensor 2, both consistent with the standard deviation of the sensor noise used to generate the reference data ($\sigma_s = 0.1$). The accuracy of the estimates is also evident in the normalised residuals shown in Fig. 9, where more than 94.5 % of the measurements from each light curve are within the ± 2 normalised residual range.

The previous results demonstrate that the measurement ambiguity can be effectively resolved through the use of stereoscopic measurements. When adopting this strategy, the AISwarm-UKF method does not incur additional computational cost, as approximately the same number of photometric measurements are used as in the single light curve scenario. In fact, the total number of particles can be reduced due to the lower presence of local optima, without compromising either the robustness in identifying the modes of the posterior probability density function or the accuracy of the final attitude estimation.

Further details on the computational performance are provided in Table 3. The AISwarm-UKF method was executed on a computer equipped with an Intel(R) Core(TM) i7-10610U CPU @ 1.80 GHz, using 8 parallel threads, and an Intel(R) UHD Graphics GPU with 7.9 GB of memory. As it can be observed, in the first test case—with 14256 particles—the most computationally demanding step is the cluster analysis, followed by the AISwarm weight

update step, which includes the simulation of measurements. In the second test case, which uses fewer than half the number of particles, the clustering time is reduced to approximately one-fifth, and the weight update step becomes the most time-consuming process. Nevertheless, the overall performance confirms that the AISwarm-UKF light curve inversion method for attitude estimation is suitable for operational application, whether using measurements from a single optical sensor or stereoscopic observations.

4. Conclusions

This study addresses the light curve inversion problem for attitude estimation. The objective is to determine the attitude state of uncontrolled space objects at a given time, assuming that their geometric and optical reflectance properties are known. The analysis is carried out with real-world operational scenarios in mind, where such estimation capabilities are particularly relevant. Examples include the development of space debris mitigation missions such as Active Debris Removal (ADR) and In-Orbit Servicing (IOS), as well as improvements in atmospheric re-entry predictions and collision risk assessments.

To this end, a novel methodology has been developed, supported by a comprehensive review of the existing literature. A critical analysis of the advantages and limitations of previously proposed methods has been instrumental in defining the objectives of the present approach, which is founded upon three key principles: exploring the complete solution space to identify all plausible solutions, including those affected by measurement ambiguities; employing batch processing of photometric measurements to mitigate local optima and facilitate parallel computation; and ensuring efficient convergence to accurate results.

Accordingly, the AISwarm-UKF method is introduced, integrating various estimation, optimisation, and data

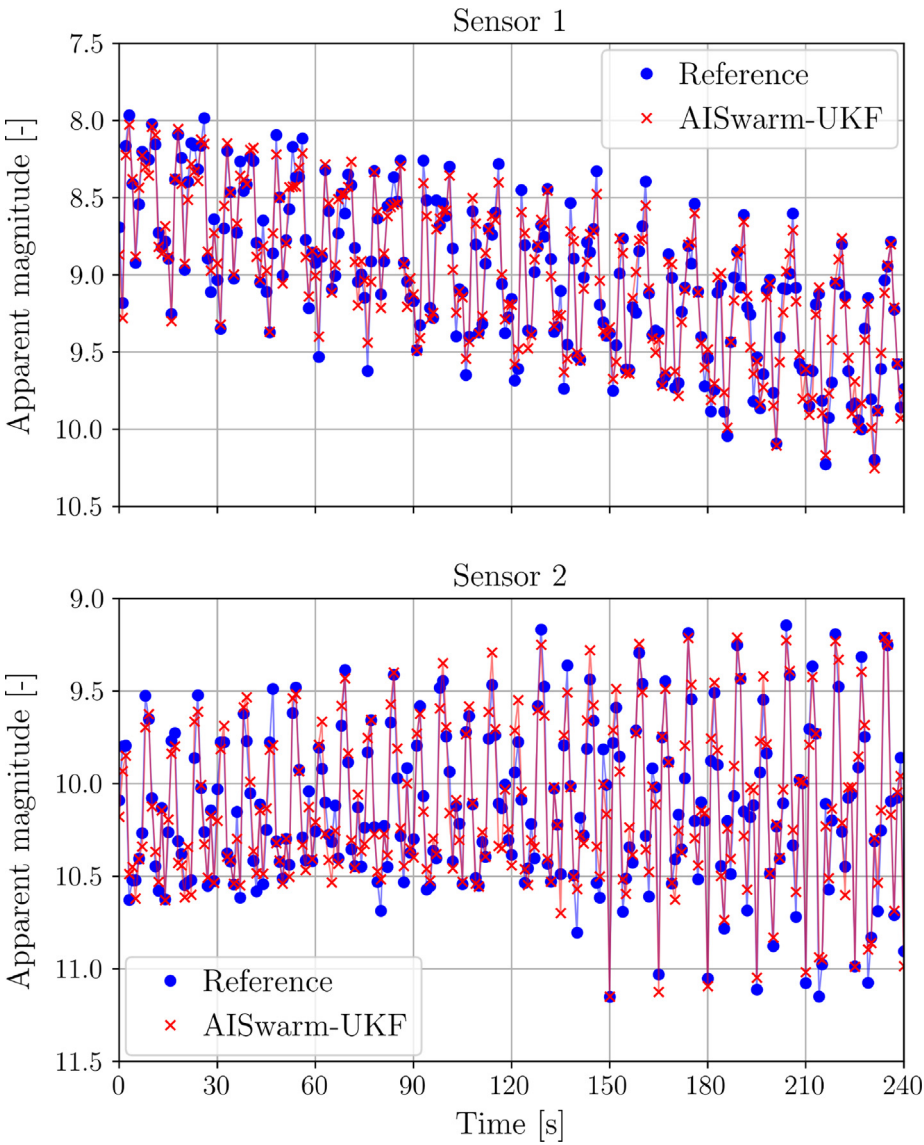


Fig. 8. Comparison of estimated and reference light curves.

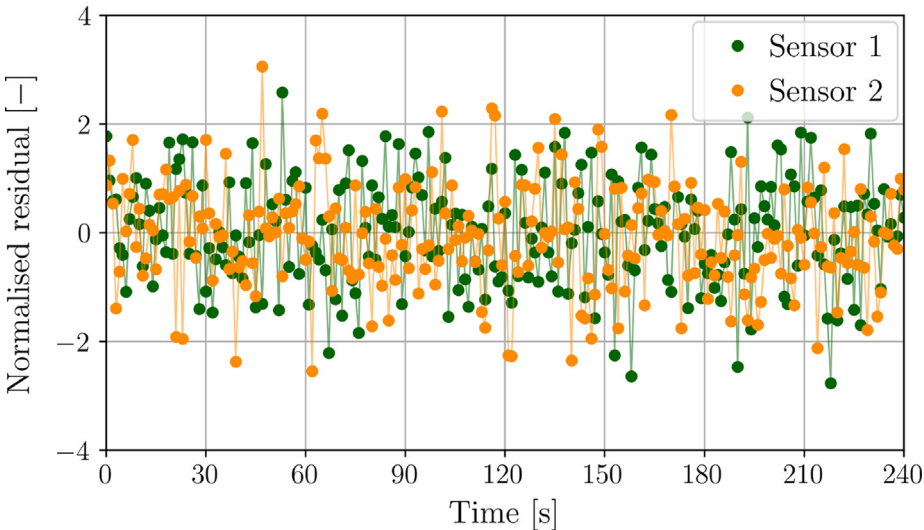


Fig. 9. Relative error between the estimated and reference light curves.

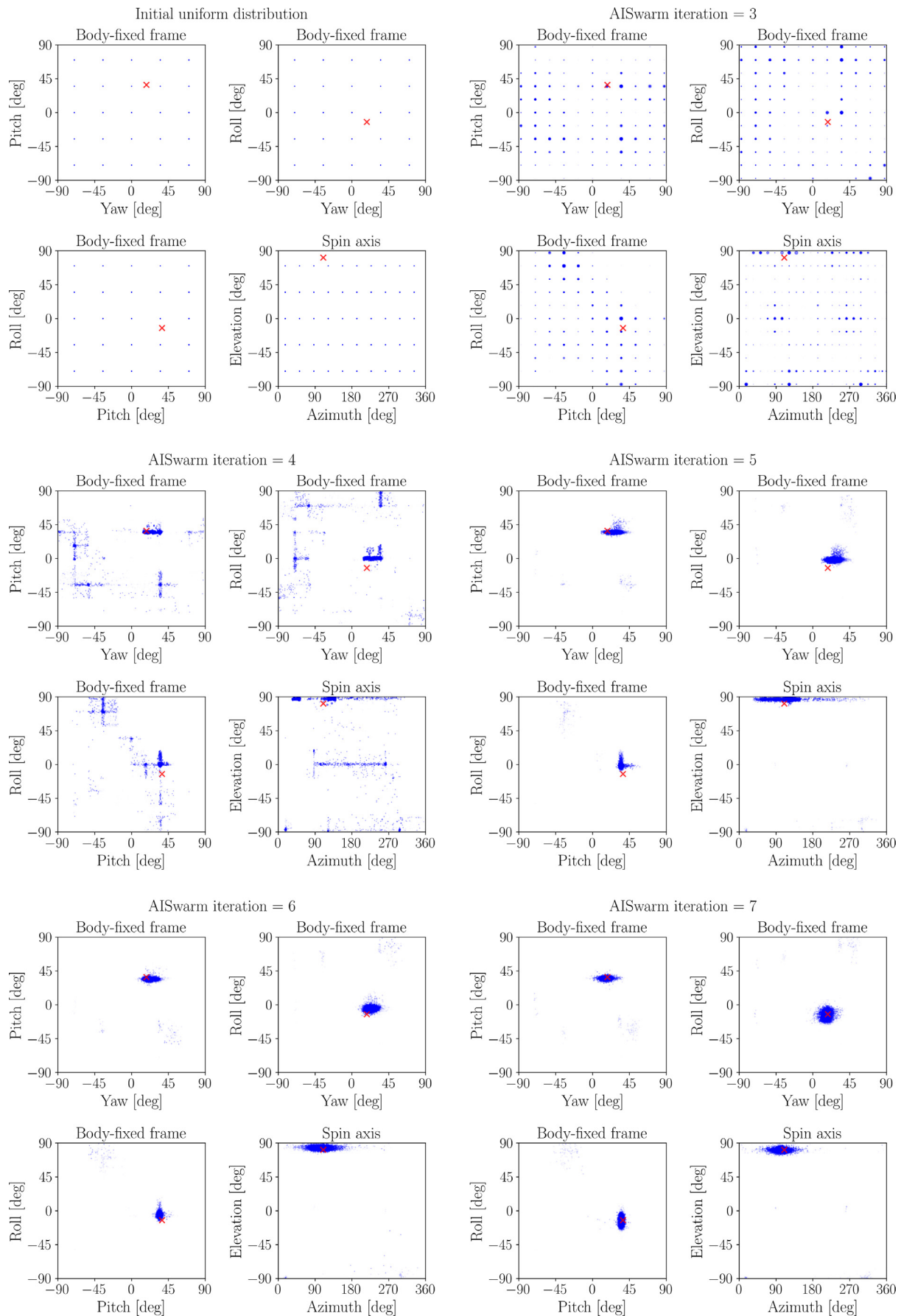


Fig. 10. AISwarm execution using stereoscopic measurements.

Table 3
Computational performance of the AISwarm-UKF method.

	Test case 1	Test case 2
Number of particles	14256	6250
Weight update (avg.) [s]	102.4	53.7
Resampling (avg.) [s]	0.07	0.03
PSO (avg.) [s]	1.6	0.6
Clustering (avg.) [s]	161.3	33.8
UKF (avg.) [s]	2.9	2.9
Total time [min]	16.2	8.3

analysis techniques: Adaptive Importance Sampling (AIS), Systematic Resampling, Particle Swarm Optimisation (PSO), Clustering, and the Unscented Kalman Filter (UKF). This method combines the strengths of each individual technique while mitigating their respective limitations through their integrated application. Specifically, the Bayesian inference process within the AIS framework benefits from the combined contribution of Systematic Resampling and PSO, which accelerates convergence towards high-probability regions. Accurate solutions are obtained through the joint effect of cluster analysis, which identifies the modes of the posterior probability density function, and the subsequent refinement of each cluster's attitude estimate using the UKF.

The performance of the AISwarm-UKF method has been evaluated using realistic test cases. The proposed attitude estimation approach successfully determines the attitude even under the most challenging conditions, such as when only a single light curve is available and no prior information about the space object's attitude is assumed. Moreover, the methodology effectively resolves measurement ambiguities through the use of stereoscopic observations, without increasing the computational load. The AISwarm-UKF method demonstrates high levels of accuracy and computational performance appropriate for space operations, especially as it is designed for computational scalability, making it well suited for integration into space surveillance missions focused on attitude estimation and monitoring of uncontrolled space objects. Nevertheless, there remains scope for further improvement. Future work may include extending the robustness analysis to a wider range of satellite models and attitude states, relaxing current assumptions about the object's optical reflectance properties and atmospheric conditions —particularly regarding the Aerosol Optical Depth (AOD)— exploring more efficient clustering algorithms, and incorporating Radar Cross-Section (RCS) measurements from ground-based radar systems.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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